



Bahir Dar University
Bahir Dar University Institute of Technology
Faculty of Computing
Requirement Analysis Document (RAD)
for
Industrial Project On GebeyaNet (AI-Powered Inventory and
SalesManagement System)

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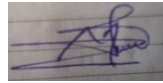
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Declaration

The project is our own work and has not been presented for a degree in any other university. All sources of materials used for this project have been duly acknowledged.

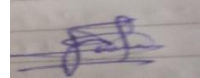
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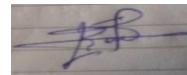
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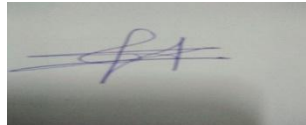
Program: Information Systems

Project Title: *GebayaNet – AI-Powered Inventory and Sales Management System*

This is to certify that I have read this project and that, in my supervision and based on the students' performance, it is fully adequate in scope and quality as a project for the award of the **Bachelor of Science in Information Systems**.

MR. Wendmu

Name of Advisor Signature:



Examining Committee Members

No.	Name	Signature	Date
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1.	Examiner 1	_____	_____
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2.	Examiner 2	_____	_____
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3.	Examiner 3	_____	_____
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It is approved that this project has been written in compliance with the formatting rules and academic standards laid down by the Faculty of Computing.

Roles and Responsibilities of the Group Members

The following matrix details the participation of each group member in key project activities, leading to the development of draft deliverables for group discussion and consensus.

Table 1: list of tasks

List of Tasks	[Member 1 Name]	[Member 2 Name]	[Member 3 Name]
Project Management & Coordination	✓		✓
Requirement Elicitation & Documentation	✓	✓	
System Analysis & UML Diagramming		✓	✓
Architectural & Database Design	✓		✓
UI/UX Prototyping & Design		✓	
Technical Writing & Document Compilation	✓	✓	✓

Acknowledgment

We extend our profound gratitude to our project advisor, [Advisor Name], for their invaluable guidance, constructive criticism, and unwavering support throughout the duration of this project. Our sincere appreciation also goes to the Faculty of Computing at the Bahir Dar Institute of Technology for providing the necessary resources and academic framework. We thank the local business owners in Bahir Dar who participated in our requirement-gathering sessions, providing essential real-world insights. Finally, we acknowledge our families and colleagues for their constant encouragement and patience.

List of Acronyms

Table 2 : list of acronyms

Acronym	Expansion
AI	Artificial Intelligence
CRC	Class-Responsibility-Collaborator
DBMS	Database Management System
ERCA	Ethiopian Revenues and Customs Authority
GUI	Graphical User Interface
KPI	Key Performance Indicator
MAPE	Mean Absolute Percentage Error
OOAD	Object-Oriented Analysis and Design
RAD	Requirement Analysis Document
RFID	Radio-Frequency Identification
SME	Small and Medium Enterprise
UI	User Interface
UML	Unified Modeling Language
POS	Point of sale

Abstract

This document provides an in-depth analysis and design for "GebayaNet" - an AI-Powered Inventory and Sales Management System designed to revolutionize business operations for retailers, wholesalers, and store owners. The system addresses critical challenges in inventory management, sales tracking, customer credit handling, and data-driven decision-making through intelligent automation and artificial intelligence integration.

GebayaNet automates stock tracking with real-time monitoring, implements predictive analytics for demand forecasting using exponential smoothing, and provides intelligent credit management with automated alerts. The system employs a customer scoring model based on payment history and purchase frequency to classify customers into loyalty tiers. The AI module analyzes sales patterns and stock data to deliver actionable business insights, forecast demand, identify fast/slow-moving products, and optimize inventory control through reorder point calculations. The system architecture follows a layered microservices approach, ensuring scalability, maintainability, and robustness. It features role-based access control for Manager, Cashier, Storekeeper, and System Administrator roles, with separate interfaces and privileges for each. The design includes comprehensive security measures, including data encryption, secure authentication, and input validation.

Key performance indicators such as Gross Profit Margin, Inventory Turnover Ratio, and Forecast Accuracy (MAPE) are calculated and visualized in real-time dashboards. The AI Chat Assistant provides natural language querying of business data, while automated reporting generates daily, weekly, and monthly performance insights.

This document serves as a comprehensive guide for the development team, detailing functional and non-functional requirements, system architecture, data models, user interfaces, and security considerations to ensure successful implementation of GebayaNet.

Chapter One: Introduction

1.1 Background

The GebayaNet AI-Powered Inventory and Sales Management System emerges as a transformative solution for Ethiopia's growing retail and wholesale sector, where manual inventory tracking, inefficient sales recording, and poor credit management continue to hinder business growth and profitability. Traditional inventory systems lack predictive capabilities, forcing business owners to rely on intuition rather than data-driven insights for restocking decisions.

In the Ethiopian context, where small and medium enterprises (SMEs) form the backbone of the economy, digital transformation in business management is crucial for competitive advantage. GebayaNet addresses this need by integrating artificial intelligence with inventory management, providing local business owners with tools previously available only to large corporations.

The system's development aligns with Bahir Dar University's commitment to fostering innovation that addresses local challenges while leveraging global technological advancements. By combining inventory management, sales tracking, credit monitoring, and AI analytics into a unified platform, GebayaNet empowers Ethiopian businesses to optimize operations, reduce losses from expiry and stockouts, and make informed decisions based on real-time data.

1.2 Statement of the Problem

The current operational paradigm for Ethiopian retailers, wholesalers, and store owners is characterized by a reliance on inefficient, manual systems that critically undermine business performance, financial health, and growth potential. This problematic environment manifests through several interconnected and critical shortcomings affecting all key actors within the business ecosystem:

- 1. Inefficient and Costly Inventory Management:** The pervasive dependency on manual, physical stock-taking results in chronic data inaccuracies, leading to disruptive stockouts or costly overstocking. Compounding this issue is the poor management of product expiry dates

due to a lack of systematic tracking, which culminates in significant financial losses from expired goods.

2. Archaic Sales and Credit Tracking Processes: Sales transactions and customer credit are predominantly recorded in physical ledgers or through disjointed notes. This handwritten approach causes operational delays, frequent errors, loss of vital records, and ineffective debt collection processes. Consequently, businesses face forgotten payments, strained customer relationships, and persistent cash flow issues.

3. The Absence of Proactive, Data-Driven Decision-Making: Critical business activities— such as restocking, procurement, and strategic planning—are conducted reactively based on intuition and guesswork. There is a severe gap in predictive capabilities and analytical insights, leaving businesses without the tools for proactive, data-driven forecasting.

4. Lack of Real-Time, Consolidated Business Intelligence: Managers and owners struggle to obtain a holistic, real-time view of business performance. The manual compilation of reports consumes valuable hours, and the absence of consolidated analytics obscures key performance metrics like profit margins, inventory turnover, and sales trends, severely hindering informed strategic decision-making.

5. Technological Accessibility and Contextual Gap: While digital solutions exist, they remain largely inaccessible to the target market. Existing software is often cost-prohibitive or lacks the necessary localization for the Ethiopian business context, language, and operational practices, creating a significant barrier to adoption.

Specific Actor-Centric Challenges:

- **Managers/Owners** struggle with the lack of real-time business performance dashboards and predictive insights.
- **Cashiers** face difficulties in efficiently processing credit sales and accurately tracking complex customer payment histories.
- **Storekeepers** cannot effectively monitor dynamic stock levels or proactively manage expiry dates across a wide product range.
- **Customers** experience service delays and frustration due to manual price lookups, uncertain stock availability checks, and cumbersome credit transaction processes.

1.3 Objectives

General Objective

To design and develop an intelligent inventory and sales management system that automates business operations,

Specific Objectives

1. Develop automated inventory tracking with real-time stock level monitoring and expiry alerts.
2. Implement sales processing with integrated cash and credit transaction handling.
3. Design an AI-powered demand forecasting system using exponential smoothing algorithms.
4. Create a customer credit scoring model with automated payment reminder systems.
5. Develop role-based interfaces for Manager, Cashier, Storekeeper, and System Administrator.
6. Implement a real-time dashboard with visual analytics and key performance indicators.
7. Integrate an AI Chat Assistant for natural language querying of business data.
8. Ensure system security through encryption, authentication, and data validation mechanisms.

1.4 Methodology

1.4.1 Requirement Gathering Methods

- **Structured Interviews:** Conduct interviews with 15-20 local shop owners, wholesalers, and retailers in Bahir Dar to understand their operational challenges.
- **Observation Studies:** Observe current manual processes in 5-10 local businesses to identify pain points and workflow inefficiencies.
- **Document Analysis:** Review existing manual record-keeping systems (ledgers, receipt books) to understand data requirements.
- **Prototype Feedback:** Develop low-fidelity prototypes and gather feedback from potential users through focus group discussions.

1.4.2 Analysis and Design Methodology

This project employs **Object-Oriented Analysis and Design (OOAD)** as its core methodology for structuring the solution. The OOAD approach was selected for its proven

efficacy in modeling complex business domains, aligning software components directly with real-world entities and workflows. By utilizing fundamental concepts such as encapsulation, inheritance, and polymorphism, this methodology ensures the development of a robust, maintainable, and adaptable system.

The primary advantages driving this choice are:

- **Enhanced Modularity:** The system is decomposed into well-defined, manageable components (e.g., Inventory Management, Sales Processing, AI Forecasting, Reporting Module). This separation of concerns simplifies development, testing, and maintenance.
- **Increased Reusability:** Common functionalities—such as User Authentication, Notification Services, and Data Validation—are designed as reusable modules. This reduces code duplication, improves consistency, and accelerates development.
- **Future Scalability:** The modular architecture establishes a clear foundation for seamless expansion. New features, such as mobile payment integration or barcode scanning, can be added in future iterations with minimal disruption to the existing codebase.

To formalize and communicate the design, the **Unified Modeling Language (UML)** serves as the standard notation for all analysis and design artifacts. For diagram generation, the project utilizes **PlantUML**, a text-based tool that ensures diagram consistency, version control compatibility, and maintainability throughout the project lifecycle.

1.4.3 Implementation Methodology

Development follows an Agile/Scrum framework with two-week sprints, enabling iterative delivery and continuous stakeholder feedback.

Process:

- **Sprints:** 2-week cycles focusing on core features like inventory and sales first.
- **CI/CD:** Daily builds and testing for quality assurance.
- **Reviews:** Bi-weekly demos with stakeholders to align development.

Technology Stack:

- **Frontend:** React.js (Web), React Native (Mobile)
- **Backend:** Node.js with Express.js
- **Database:** PostgreSQL, Redis (cache)
- **AI/ML:** Python, scikit-learn

- **Deployment:** The application is deployed using free-tier cloud platforms, with the frontend hosted on Vercel and the backend hosted on Render.

1.5 Feasibility

Technical Feasibility

The project leverages established and widely supported technologies, including React, Node.js, PostgreSQL, and Python. The team possesses foundational knowledge in these areas, and ample documentation, tutorials, and community resources are available to support development. The proposed architecture is modular and can be deployed using scalable cloud infrastructure with accessible free-tier options, ensuring the technical goals are achievable.

Economic Feasibility

The development and operational costs are minimal and sustainable. Primary expenses are limited to cloud service subscriptions during development and a modest monthly hosting fee for production deployment. The system presents a clear value proposition for Ethiopian retailers. A tiered subscription model is proposed, which is projected to generate significant annual revenue with a relatively small customer base, ensuring a strong return on investment and long-term economic viability.

Time Feasibility

A realistic and structured six-month development timeline has been outlined. This period is segmented into dedicated phases for requirements gathering, core system development, AI module integration, thorough testing, and final deployment. This phased approach allows for focused effort, regular milestone completion, and manageable workload distribution, making the proposed schedule feasible for the development team.

1.6 Beneficiaries

Primary Beneficiaries

Small and medium-sized Ethiopian retailers, wholesalers, and store owners are the primary beneficiaries. They will gain direct operational and financial advantages, including automated inventory control to reduce losses from expiry and stockouts, efficient sales and

credit management to improve cash flow, and data-driven dashboards to enable informed decision making and strategic planning.

Secondary Beneficiaries

The benefits of the system extend to the broader business ecosystem. Customers will experience faster and more reliable service with accurate product information. Suppliers can anticipate more predictable ordering patterns, leading to better supply chain planning. Financial institutions could leverage transactional data from the system to develop credit scoring models that support responsible lending to small businesses.

Significance and Broader Impact

This project holds significance beyond its immediate users. It represents a meaningful step toward the digitalization of Ethiopia's vital retail sector, fostering greater operational resilience. By promoting the adoption of accessible technology, it contributes to digital literacy among traditional business owners. The development process itself builds local technical capacity. Furthermore, the project serves as a practical case study for implementing scalable, AI-enhanced solutions in resource-constrained environments, with the potential to stimulate job creation in technical support and contribute to broader local economic development.

1.7 Limitations

The project acknowledges the following constraints and operational conditions that define the scope of the initial release:

1. **Data Dependency for AI:** The accuracy of the predictive forecasting module is contingent on the availability of sufficient historical transaction data. A business will need at least three to six months of consistent sales data logged within the system to generate reliable insights.
2. **Connectivity Requirement:** Core functionalities involving real-time data synchronization, multi-user access, and cloud backup require a consistent and stable internet connection. Offline capabilities will be limited to basic transaction recording with synchronization upon reconnection.
3. **Language Scope:** The initial software release will provide user interfaces in English and Amharic only, targeting the primary linguistic contexts of the intended user base.

4. **Deferred Features:** To prioritize core inventory and sales management, certain advanced features are planned for subsequent development phases. These include direct API integration with mobile money platforms (requiring initial manual payment entry) and native support for hardware barcode scanners, which will require compatible external hardware.

1.8 Scope

Scope

The project scope is defined to ensure a focused and viable initial release.

In-Scope (Phase 1)

This phase will deliver the complete core management system, encompassing:

1. **Inventory Management:** Full CRUD (Create, Read, Update, Delete) operations, stock level tracking, and expiry date monitoring.
2. **Sales Processing:** Handling of both cash and credit transactions with receipt generation.
3. **Customer & Supplier Management:** Maintaining records, tracking customer credit history with basic scoring, and managing supplier information with purchase order tracking.
4. **AI-Driven Forecasting:** A module for inventory and sales demand prediction using machine learning.
5. **Reporting & Analytics:** A real-time business intelligence dashboard with key performance indicators and automated report generation.
6. **User Management:** A role-based access control system for four distinct user roles (e.g., Manager, Cashier, Storekeeper).
7. **Client Applications:** A web-based administration panel and a companion mobile application for on-floor operations.
8. **Automated Notifications:** System-generated alerts for low stock, approaching expiry dates, and overdue customer credit.

Out-of-Scope (Deferred for Future Phases)

The following advanced features are explicitly excluded from the initial release to maintain focus but are considered for subsequent development:

1. **Integrated Payment Gateways:** Direct API integration with mobile money or banking platforms.
2. **Hardware Integration:** Native support for barcode scanners or RFID readers.
3. **Advanced Inventory:** Multi-location or multi-warehouse inventory synchronization.

4. **Advanced Analytics:** Predictive models utilizing deep learning techniques.
5. **External System Integration:** Direct data exchange with government tax or regulatory systems.

1.9 Organization of the Project

This document is structured as follows: Chapter One provides the project's introduction, objectives, and methodology. Chapter Two details the system features, requirements analysis, and models. The subsequent design specification, including architecture and database schema, will be covered in a separate System Design Document.

Chapter Two: System Features

2.1 The Existing System

The prevailing operational model among target Ethiopian SMEs is characterized by fundamentally manual, low-tech processes. Inventory management is typically conducted through infrequent physical stock counts. Sales transactions, both cash and credit, are recorded in paper ledgers or basic, non-automated spreadsheet files. Customer credit information is often maintained in separate notebooks or informal records.

This approach creates a **fragmented and inefficient ecosystem** where inventory, sales, and financial data exist in isolation. The consequences are systemic: chronic data inconsistency, a high propensity for human error, an inability to access real-time business insights or generate consolidated reports, and a complete absence of predictive capabilities for planning. These limitations directly impede operational efficiency, obscure financial performance, and constrain strategic decision-making and growth.

2.2 The Proposed System: GebayaNet

GebayaNet is a centralized, intelligent business management platform designed to replace the fragmented, manual systems currently in use. The proposed system will deliver an integrated solution that provides real-time visibility and control over all critical operations.

The core functionality includes:

- **Real-Time Inventory Management:** Automated tracking of stock levels, with instant updates across sales and procurement.
- **Intelligent Sales & Credit Processing:** Streamlined handling of cash and credit transactions, supported by automated customer credit scoring and payment reminders.
- **Predictive Business Insights:** AI-driven demand forecasting models that analyze historical sales data to inform restocking and planning.
- **Unified Operational Dashboard:** A consolidated view of key performance indicators, inventory status, sales trends, and financial summaries.
- **Role-Based Accessibility:** Tailored web and mobile interfaces providing customized workflows and information access for managers, cashiers, storekeepers, and system administrators.

2.3 Requirement Analysis

2.3.1 Functional Requirements Analysis

TABLE 3: Functional Requirements Analysis

ID	Requirement	Description	Priority	ID
FREQ-1	Inventory Management	Add, update, delete products with attributes (name, category, price, quantity, expiry)	High	FREQ-1
FREQ-2	Sales Processing	Record cash and credit sales with receipt generation	High	FREQ-2
FREQ-3	Stock Monitoring	Real-time tracking of stock levels with lowstock alerts	High	FREQ-3
FREQ-4	Expiry Tracking	Monitor product expiry dates with advance notifications	High	FREQ-4
FREQ-5	Customer Credit Management	Record customer details, track credits, send payment reminders	High	FREQ-5
FREQ-6	Demand Forecasting	Predict product demand using	Medium	FREQ-6
		historical sales data		
FREQ-7	Reporting Dashboard	Generate sales, inventory, and profit reports	High	FREQ-7
FREQ-8	Role-Based Access	Different interfaces for Manager, Cashier, Storekeeper, Admin	High	FREQ-8
FREQ-9	AI Chat Assistant	Natural language queries about business data	Medium	FREQ-9
FREQ-10	Supplier Management	Maintain supplier records and purchase history	Medium	FREQ-10

2.3.2 System Use Case

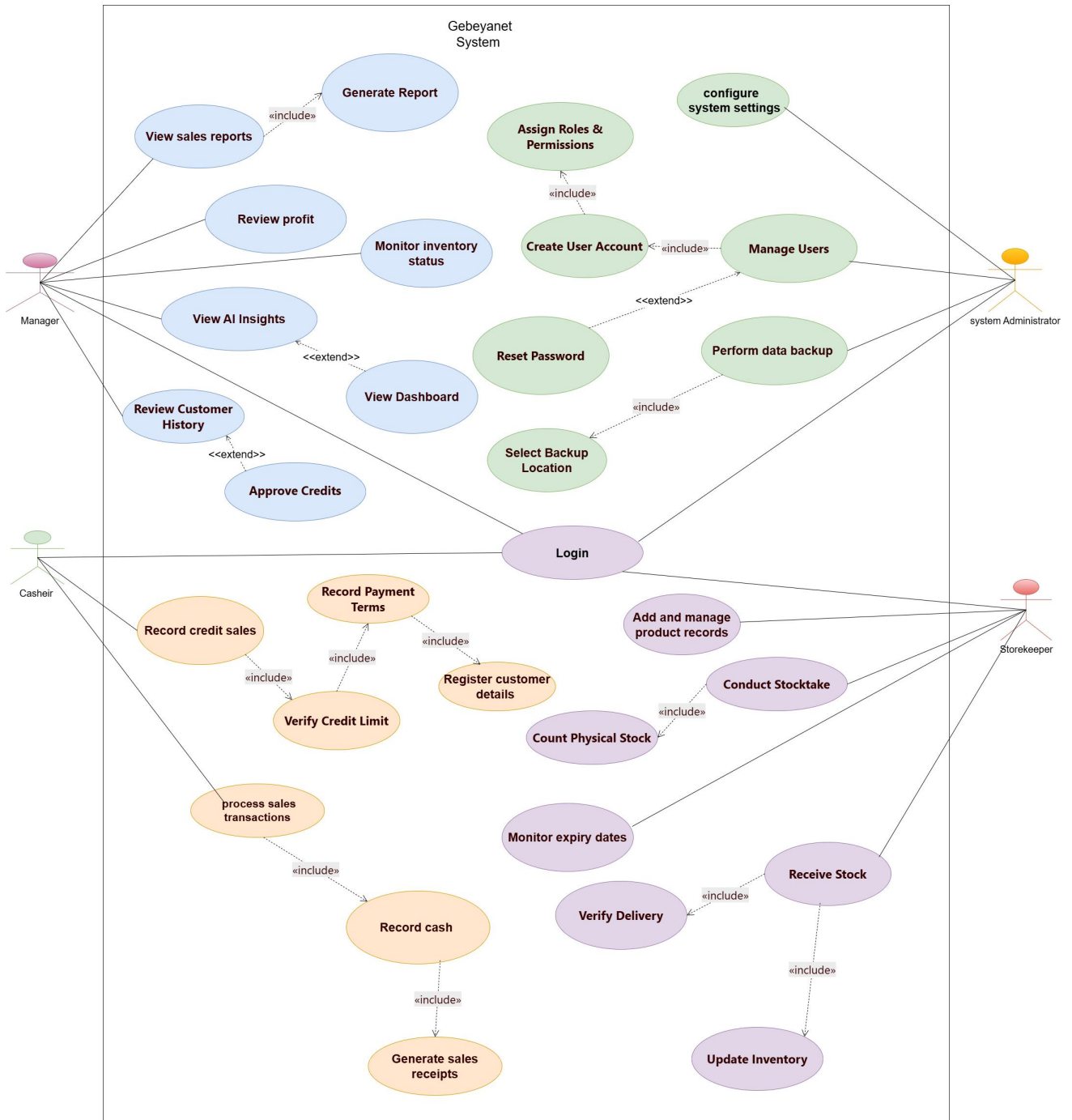


Figure 1 Use Case Diagram

2.3.2.2 Use Case Documentation

This section presents detailed descriptions of the system use cases identified in the use case diagram. Each use case is documented by specifying its identifier, name, actors, description, preconditions, postconditions, priority, basic course of action, and alternate course of action.

Use Case UC-01: Login to System

Primary Actor:

Manager, Cashier, Storekeeper, System Administrator

Description:

This use case allows an authorized user to log into the GebayaNet system using valid credentials in order to access system functionalities based on assigned roles.

Pre-condition:

- The user is registered in the system.
- The system is available and online.

Post-condition:

- The user is successfully authenticated.
- The user dashboard is displayed according to their role.

Priority: High

Table 4: Login to System

Step	Actor Action (UI)	System Business Rule	Response /
1	User opens the login page	System displays login interface	
2	User enters username and password	System validates input format	
3	User clicks the Login button	System verifies credentials	
4	—	System grants access and redirects user to dashboard	

Alternate Course of Action

Step	Condition	System Response
2a	Username or password is empty	System displays validation error
3a	Invalid credentials	System denies access and shows error message

Use Case UC-02: Manage Products

Primary Actor:

Manager, Storekeeper

Description:

This use case allows authorized users to add, update, view, or remove product information from the system.

Pre-condition:

- The user is logged into the system.
- The user has product management permission.

Post-condition:

- Product information is updated in the database.

Priority: High

Basic Course of Action

Table 5: Manage Products

Step	Actor Action (UI)	System Response / Business Rule
1	User opens product management page	System displays product list
2	User selects add/edit/delete option	System shows product form
3	User enters product details	System validates required fields
4	User saves changes	System updates product
		records

Alternate Course of Action

Step	Condition	System Response
3a	Required fields missing	System displays error message
4a	Duplicate product detected	System prevents saving

Use Case UC-03: Process Sale

Primary Actor:

Cashier

Description:

This use case enables the cashier to record a cash sale transaction for purchased products.

Pre-condition:

- The cashier is logged into the system.
- Products are available in stock.

Post-condition:

- The sale transaction is recorded.
- The inventory quantity is updated.

Priority: High

Table 6: Process Sale Basic Course of Action

Step	Actor Action (UI)	System Response / Business Rule	Step
1	Cashier opens sales page	System displays sales interface	1
2	Cashier selects products	System calculates total price	2
3	Cashier confirms payment	System records sale transaction	3
4	—	System updates inventory levels	4

Alternate Course of Action

Step	Condition	System Response
2a	Product out of stock	System prevents selection
3a	Transaction cancelled	System discards sale

Use Case UC-04: Process Credit Sale

Primary Actor:

Cashier

Description:

This use case allows the cashier to record a credit sale by capturing customer identification details and repayment deadline.

Pre-condition:

- The cashier is logged in.
- Customer information is available.

Post-condition:

- The credit sale is recorded.
- The repayment deadline is scheduled.

Priority: High

Basic Course of Action

Table 7: Process Credit

Step	Actor Action (UI)	System Response / Business Rule
1	Cashier selects credit sale option	System displays credit form
2	Cashier enters customer ID and phone	System validates customer data
3	Cashier sets repayment date	System records credit transaction
4	—	System schedules reminder alert

Alternate Course of Action

Step	Condition	System Response
2a	Invalid customer details	System displays error
3a	Repayment date missing	System blocks submission

Use Case UC-05: Manage Inventory Stock

Primary Actor:

Storekeeper

Description:

This use case allows the storekeeper to monitor stock levels and update inventory records.

Pre-condition:

- The storekeeper is logged in.

Post-condition:

- The inventory records are updated.

Priority: High**Basic Course of Action**

Table 8: Manage Inventory Stock

Step	Actor Action (UI)	System Response / Business Rule
1	Storekeeper opens inventory page	System displays stock levels
2	Storekeeper updates quantities	System validates stock values
3	Storekeeper saves changes	System updates inventory database

Alternate Course of Action

Step	Condition	System Response
2a	Invalid quantity entered	System rejects input

Use Case UC-06: Generate Reports**Primary Actor:**

Manager

Description:

This use case enables the manager to generate sales and inventory reports for analysis and decision-making.

Pre-condition:

- Manager is logged into the system
- Transaction data exists

Post-condition:

- Report is generated and displayed

Priority: Medium

Basic Course of Action

Table 9: Generate Reports

	Actor Action (UI)	System Response / Business Rule
1	Manager opens reports page	System displays report options
2	Manager selects report type	System processes data
3	Manager generates report	System displays report

Alternate Course of Action

Step	Condition	System Response
2a	No data available	System shows notification

Use Case UC-07: Forecast Product Demand (AI)

Primary Actor:

Manager

Description:

This use case allows the manager to view AI-generated demand forecasts based on historical sales data.

Pre-condition:

- Manager is logged in
- Sales data is available

Post-condition:

Demand forecast is displayed

Priority: Medium

Basic Course of Action

Table 10: Forecast Product Demand (AI)

Step	Actor Action (UI)	System Response / Business Rule
1	Manager opens AI analytics page	System analyzes sales data
2	Manager requests forecast	AI module generates prediction
3	—	System displays forecast results

Alternate Course of Action

Step	Condition	System Response
1a	Insufficient data	System displays warning

2.3.3 Business Rule Documentation

This section describes the **rules and constraints** that govern the operation of the GebayaNet system. These business rules ensure data accuracy, security, consistency, and proper system behavior.

BR-01 User Authentication Rule

- Only registered users with valid credentials shall be allowed to access the system.
- Access rights shall be granted based on the user's assigned role.
- Unauthorized access attempts shall be denied.

Applies to: Manager, Cashier, Storekeeper, System Administrator

BR-02 Role-Based Access Control Rule

- Each user shall access only the functions permitted to their role.
- Cashiers shall not manage users or system settings.
- Storekeepers shall not process sales transactions.
- Only the System Administrator shall manage users and roles.

Applies to: All internal users

BR-03 Product Management Rule

- Product name, category, price, and quantity are mandatory fields.
- Duplicate products with the same name and category shall not be allowed.
- Product price must be greater than zero.

Applies to: Manager, Storekeeper

BR-04 Inventory Stock Rule

- Inventory quantity shall automatically decrease after every successful sale.
- Inventory quantity shall increase after supplier delivery confirmation.
- Stock quantity shall never be less than zero.

Applies to: Storekeeper, Cashier

BR-05 Low Stock Alert Rule

- The system shall generate an alert when product quantity falls below the defined threshold.
- Low stock alerts shall be visible on the dashboard. **Applies to:** Manager, Storekeeper

BR-06 Product Expiry Rule

- Products with expiry dates shall be monitored daily.
- The system shall alert users before a product reaches its expiry date.
- Expired products shall not be allowed for sale.

Applies to: Storekeeper, Cashier

BR-07 Sales Transaction Rule

- A sale transaction shall be recorded only if the product is in stock.
- Each sale transaction shall include date, items, quantity, and total amount.
- Cancelled transactions shall not affect inventory. **Applies to:** Cashier

BR-08 Credit Sale Rule

- Credit sales shall require customer identification and phone number.
- A repayment deadline must be specified for every credit sale.
- Credit reminders shall be generated before the due date.

Applies to: Cashier, Manager

BR-09 Customer Credit Classification Rule

- Customers shall be classified as loyal, occasional, or high-risk based on repayment history.
- High-risk customers may be restricted from further credit sales.

Applies to: Manager

BR-10 AI Forecasting Rule

- AI demand forecasting shall use historical sales data.
- Forecast results shall be advisory and not automatically executed.
- Forecasts shall be updated periodically.

Applies to: Manager

BR-11 Reporting Rule

- Reports shall be generated only from completed transactions.
- Reports shall be available in daily, weekly, and monthly formats. **Applies to:** Manager

BR-12 Data Backup Rule

- System data shall be backed up automatically to cloud storage.
- Only authorized users shall restore backup data.

Applies to: System Administrator

2.3.4 User Interface Prototype

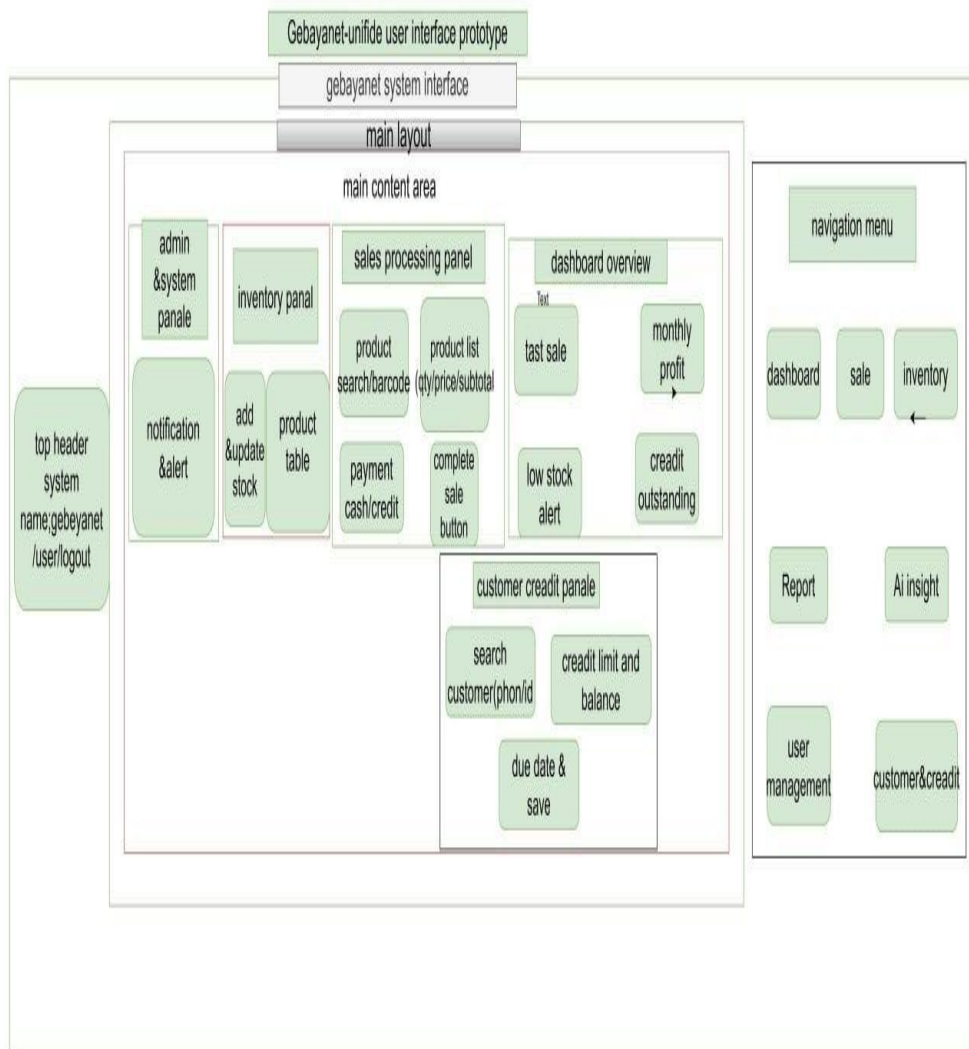


Figure 2 *user interface prototype*

2.3.5 State Chart Diagram

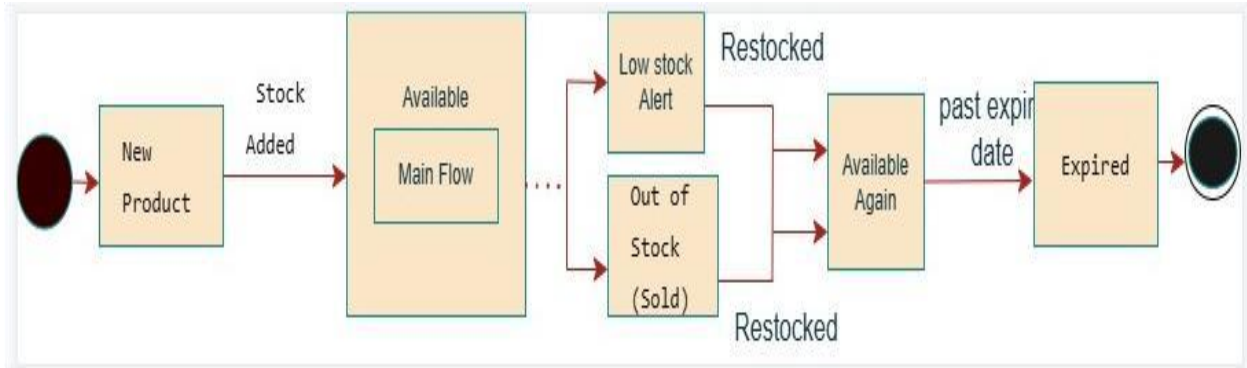


Figure 3 Product State Chart Diagram

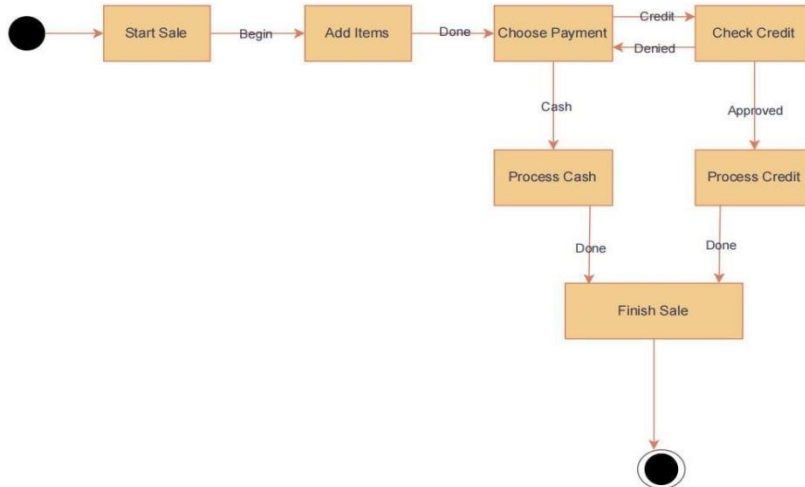


Figure 4 Sales State Chart

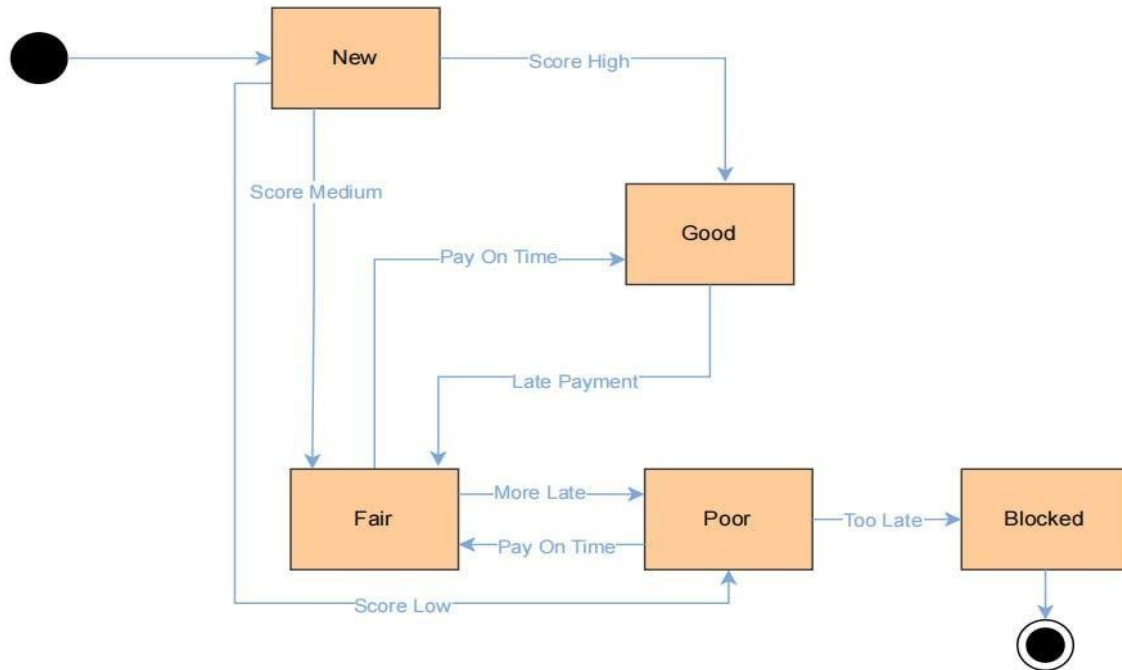


Figure 5 Customer Credit State

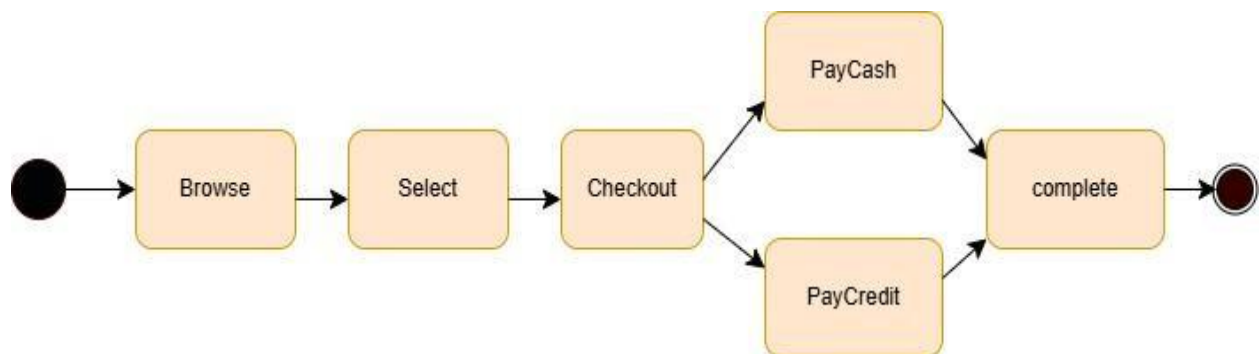


Figure 6 **Purchase Flow** state chart diagram

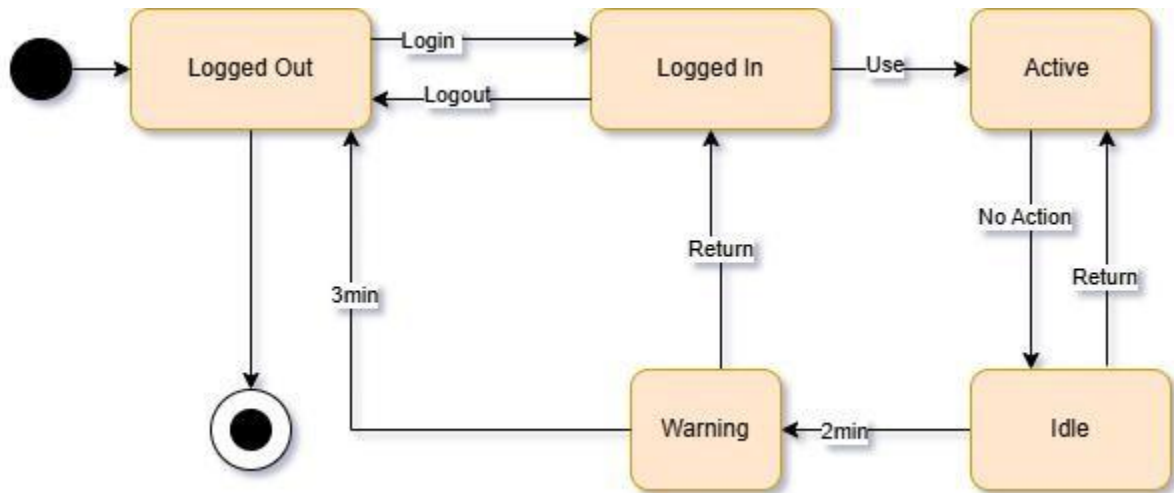


Figure 7 **User session state chart diagram**

2.3.6 Activity Diagram

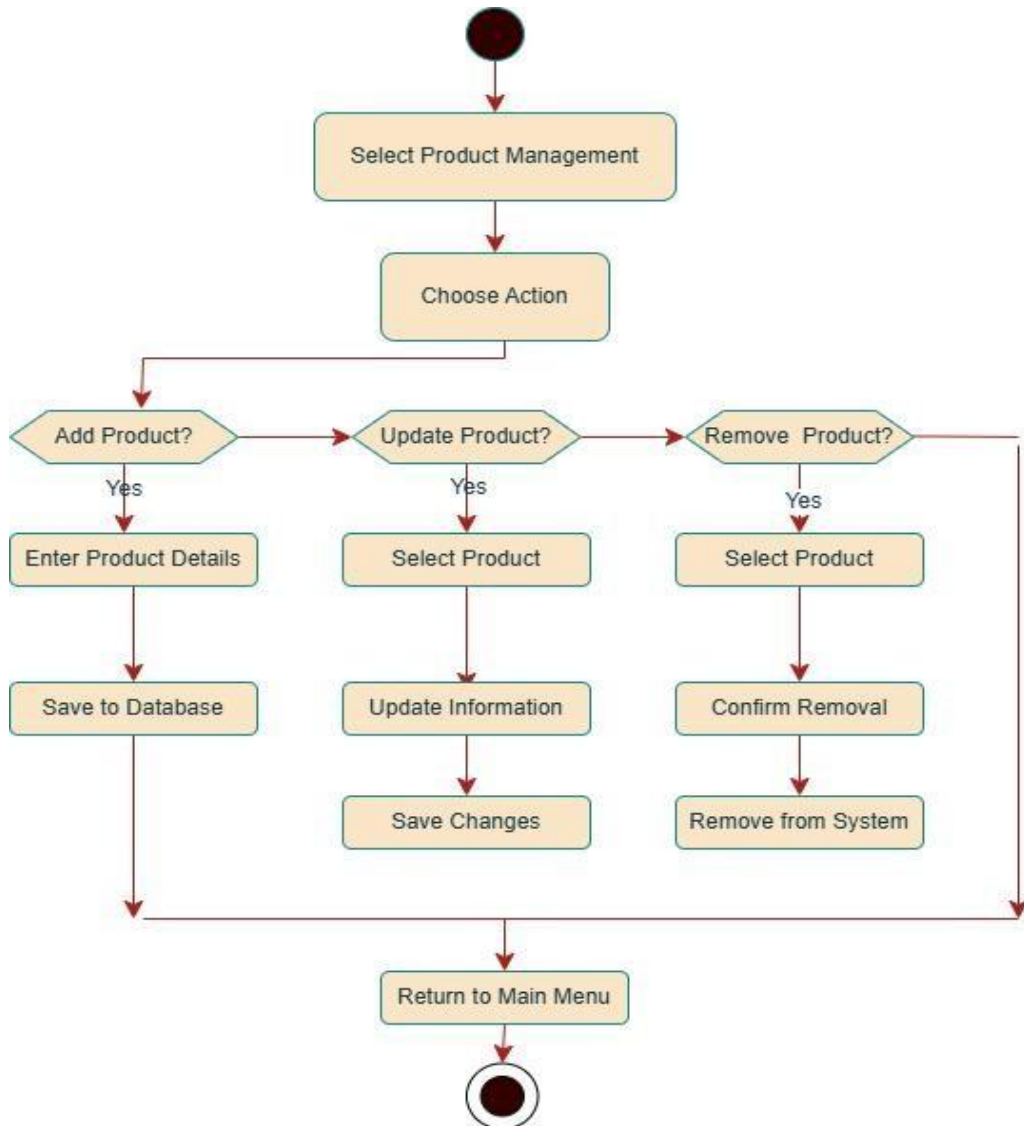


Figure 8 *Product management activity diagram*

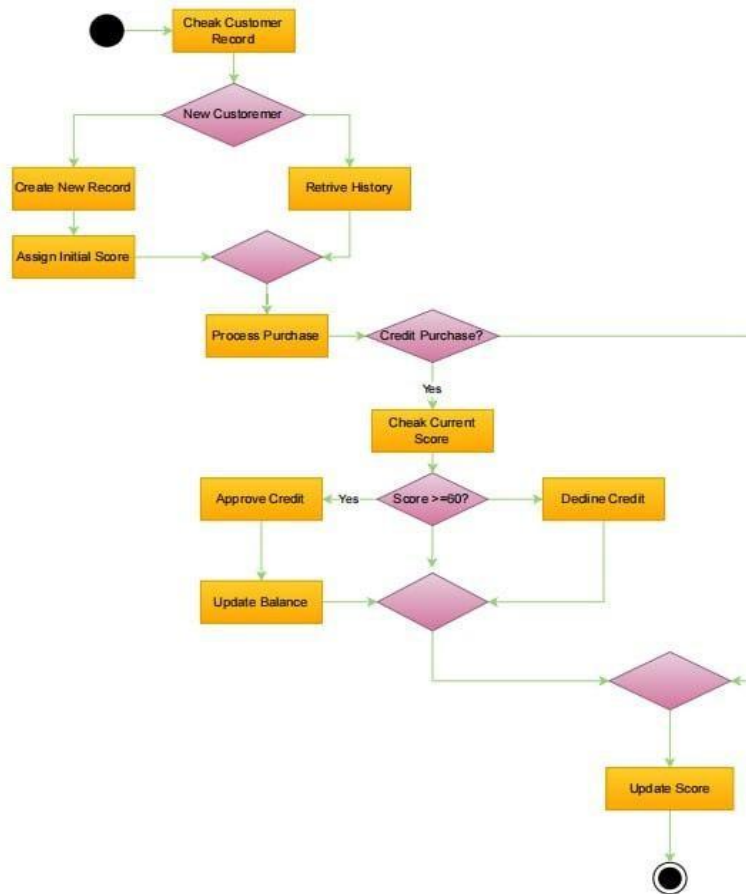


Figure 9 **Customer credit activity Diagram**

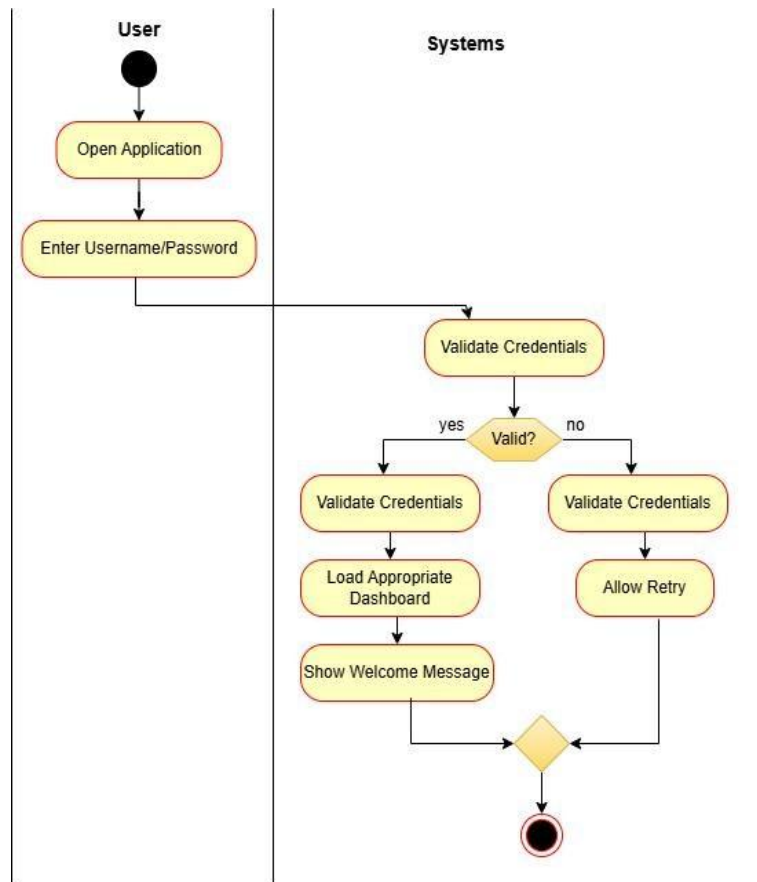


Figure 10 *User login activity Diagram*



Figure 11 *Daily sales activity diagram*

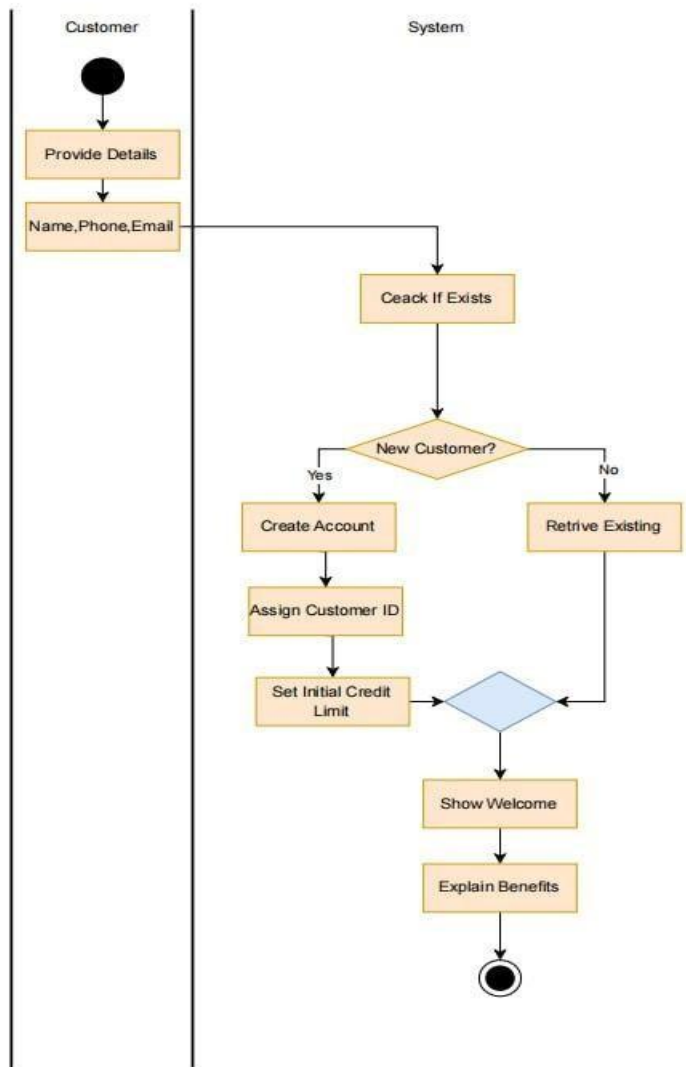


Figure 12 Customer registration activity

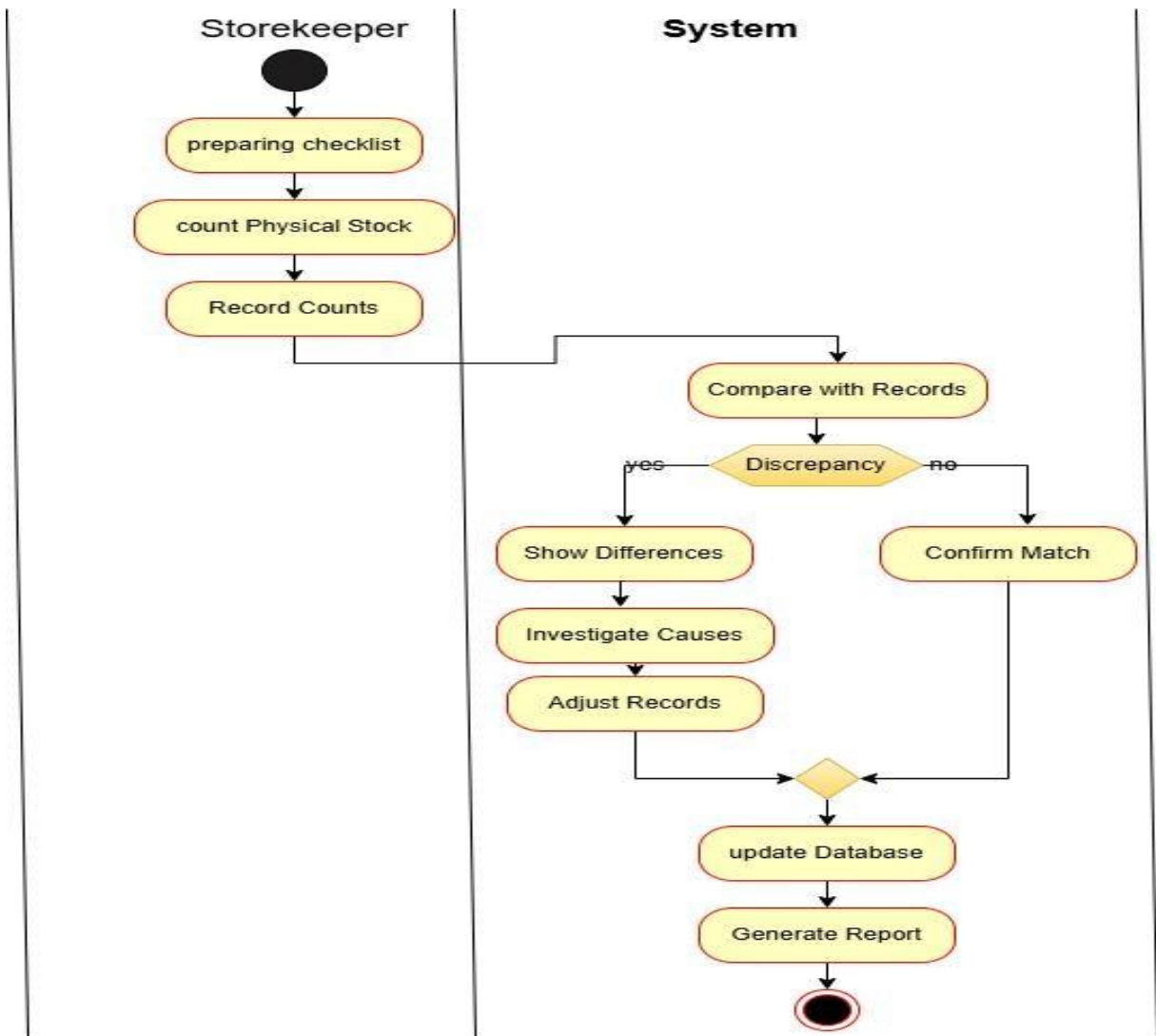


Figure 13 *stock taking activity*

2.3.7 Sequence Diagram

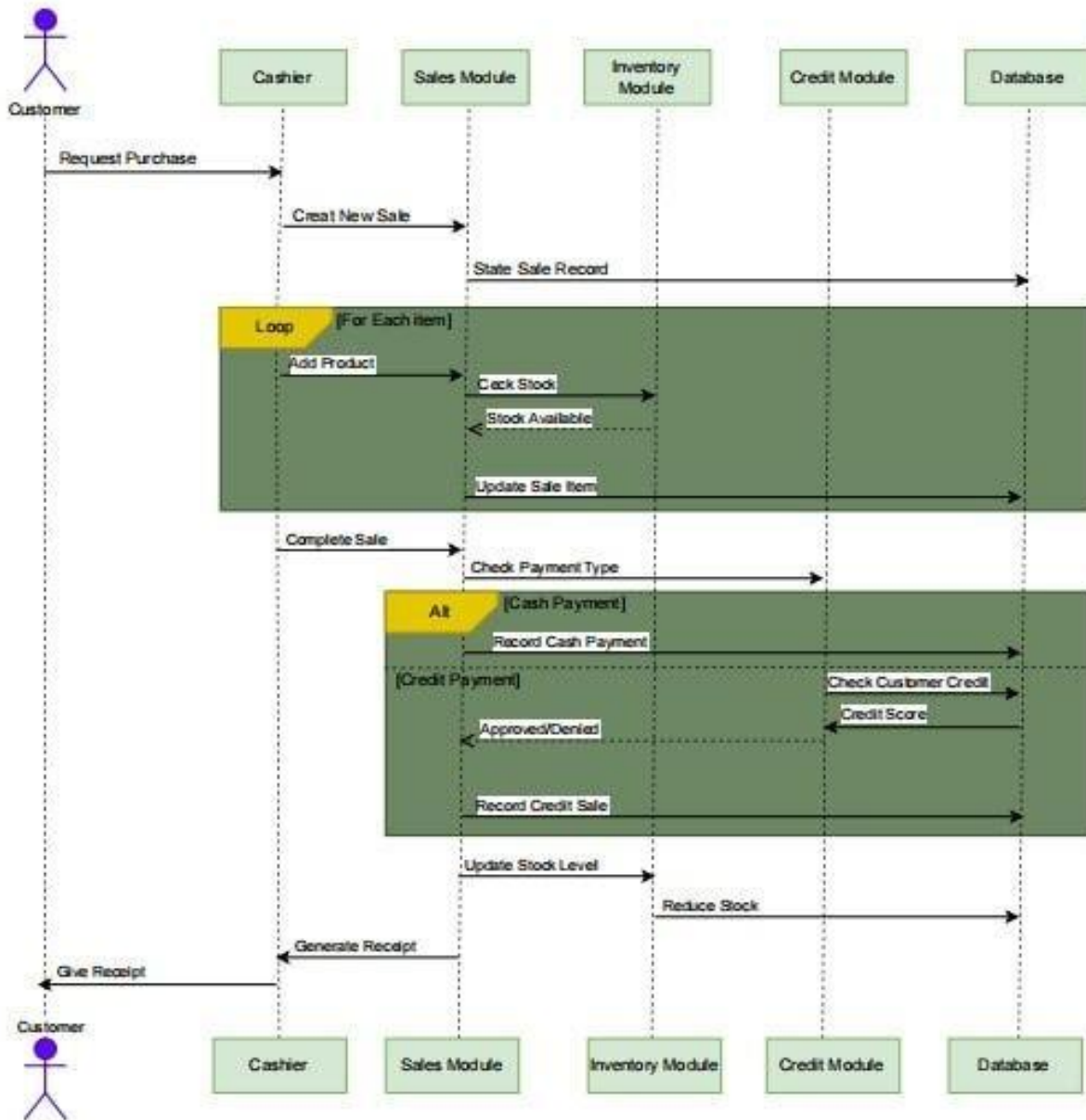


Figure 14 Sales Processing Sequence

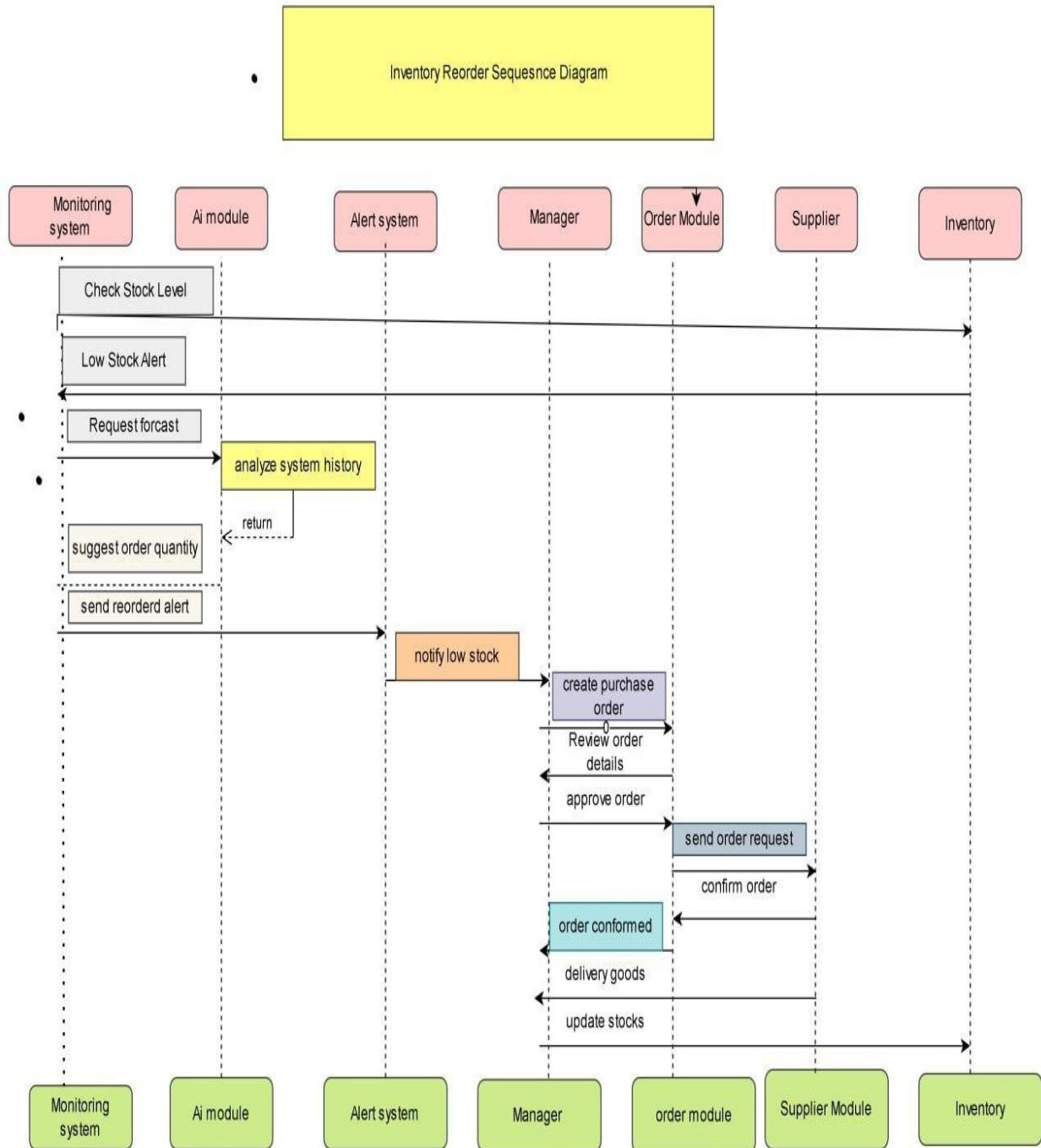


Figure 15 Inventory Reorder Sequence

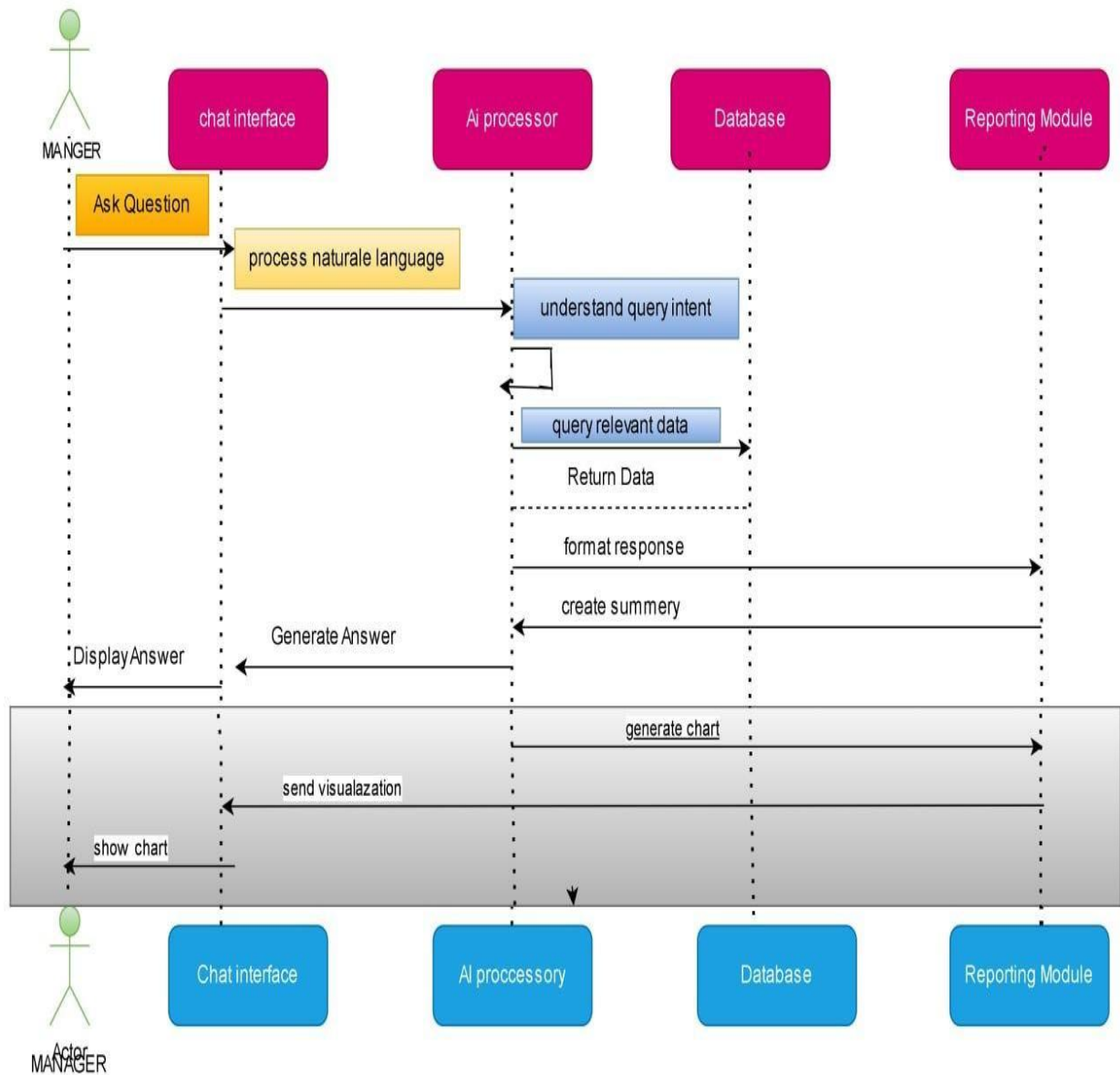


Figure 16 **AI Assistant Query Sequence**

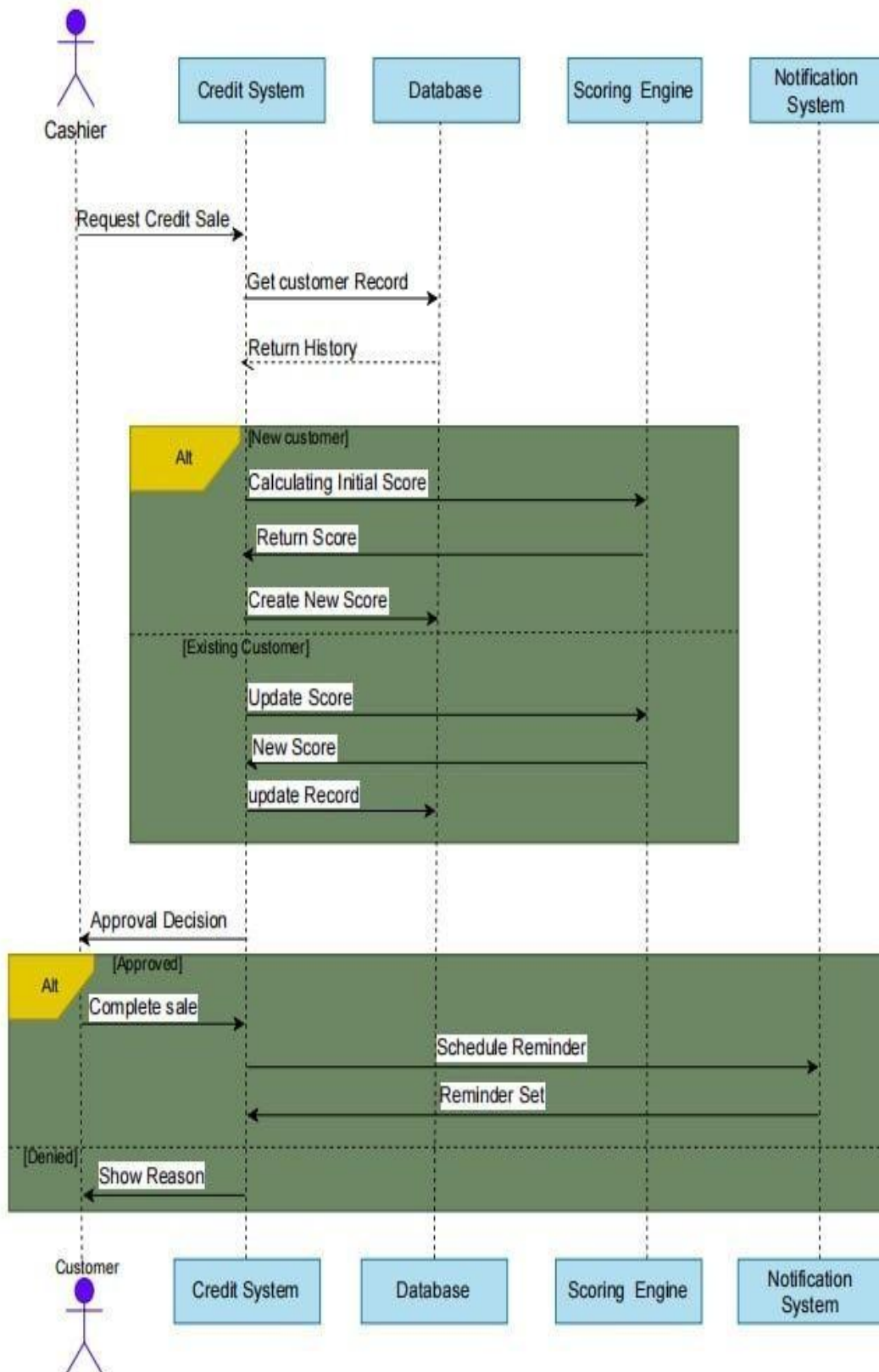


Figure 17 Customer Credit Check Sequence4

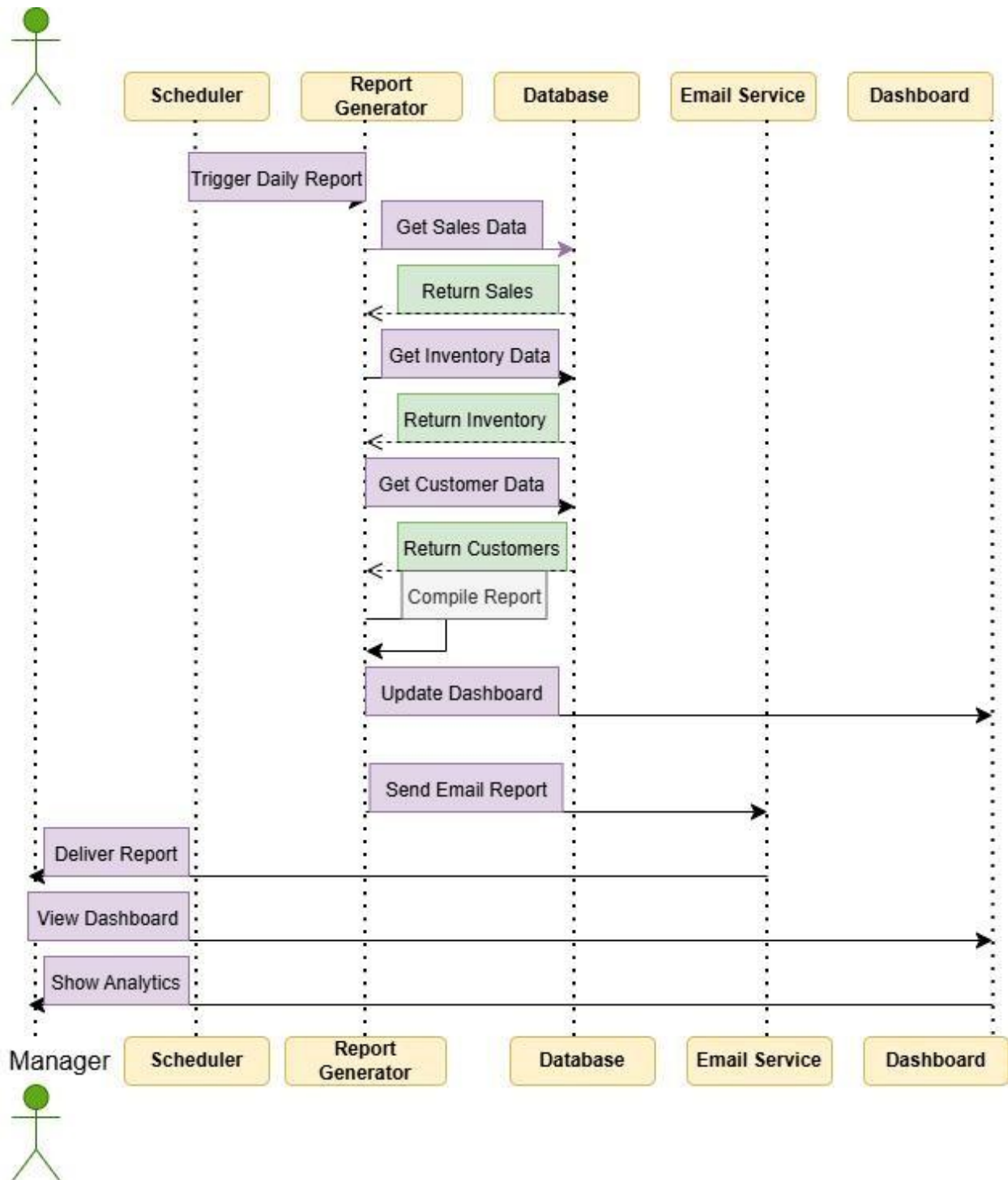


Figure 18 Daily report generation

User Login Sequence diagram

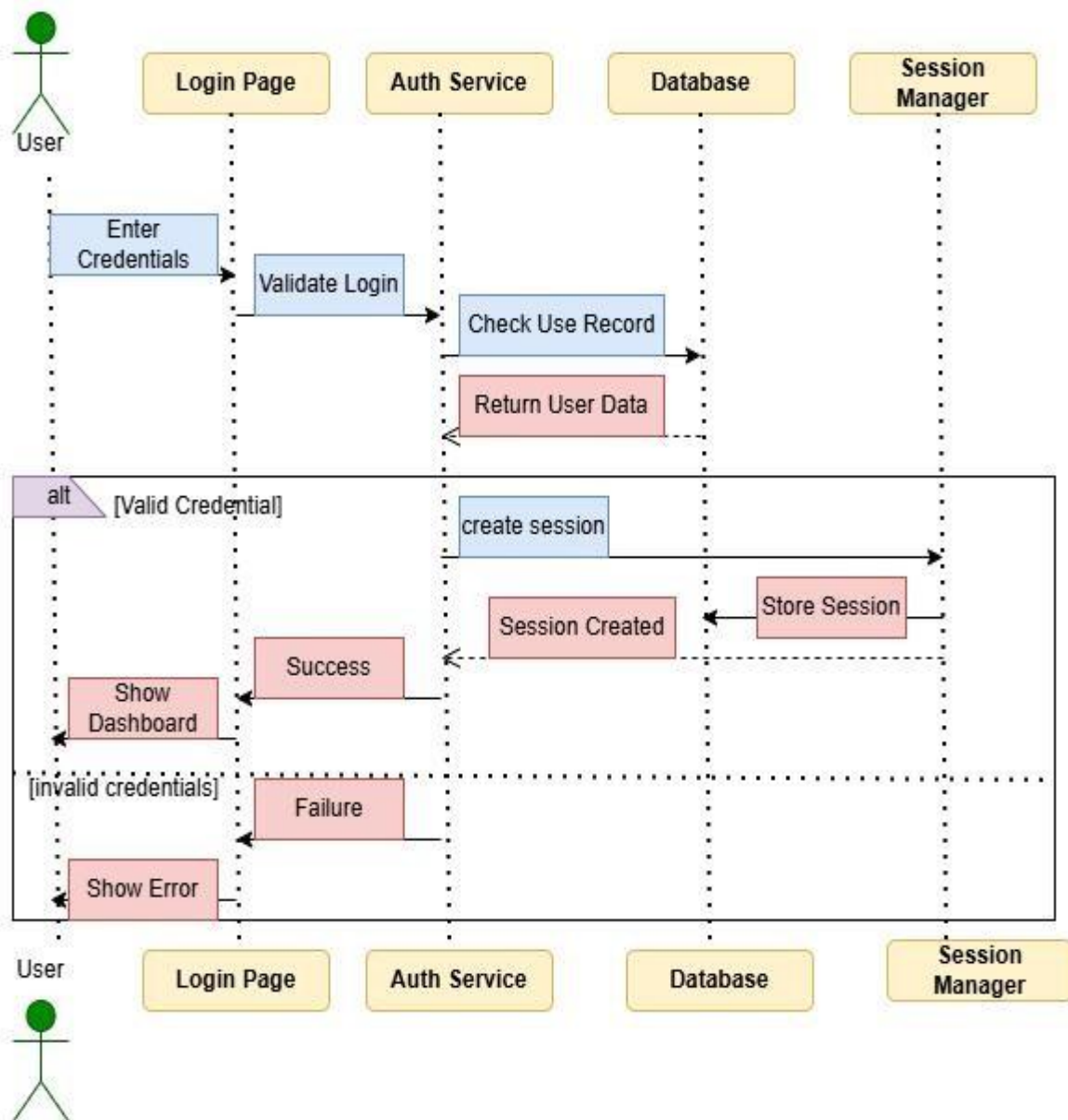


Figure 19 User Login Sequence Diagram

2.3.8 Analysis Class Model

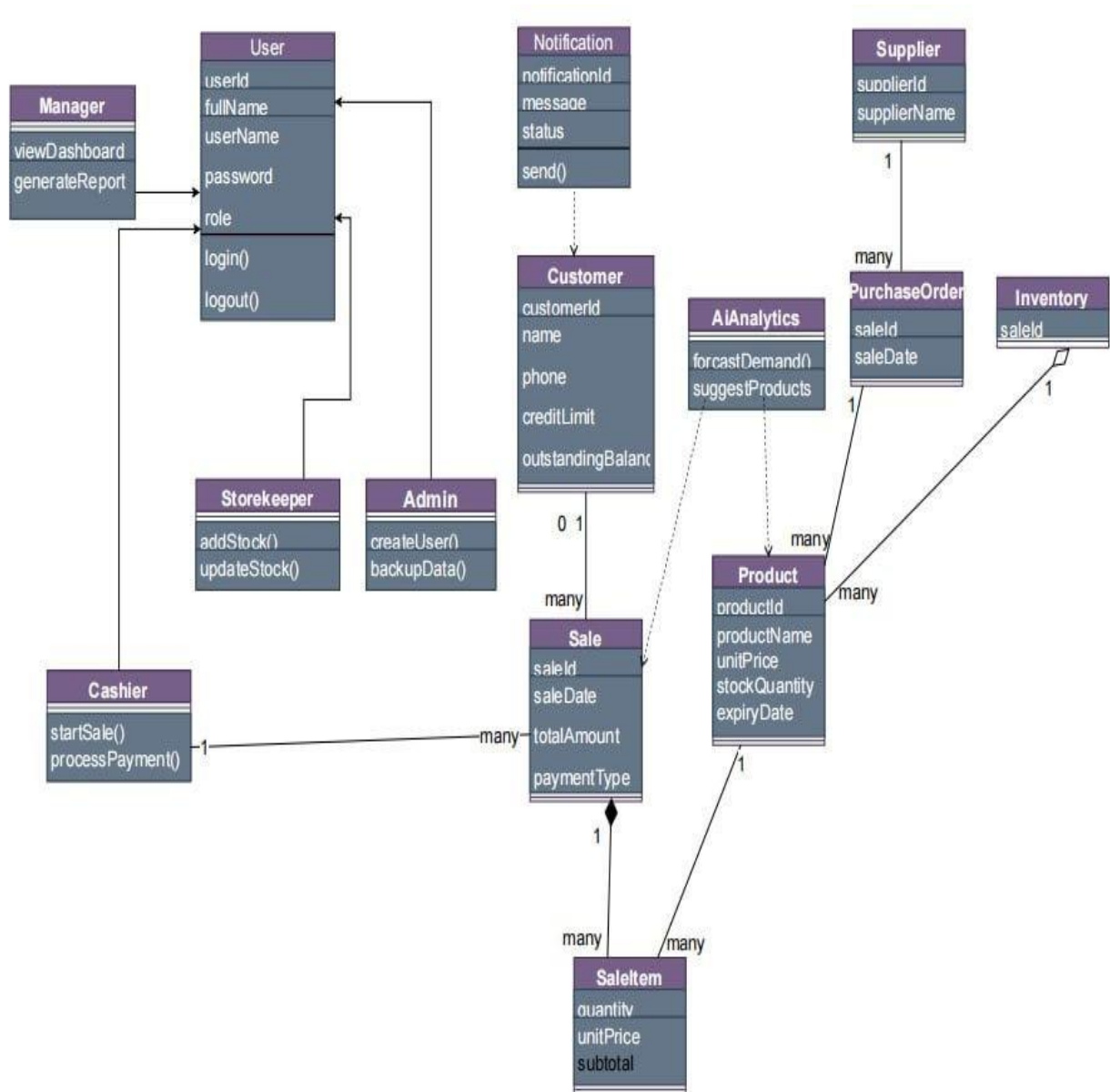


Figure 20 **Analysis class model**

2.3.9 Logic Mode

The logic model describes the **step-by-step logical flow** of the major processes of the GebayaNet system. Each algorithm explains how inputs are processed to produce expected outputs.

Logic Model 1: Daily Sales Processing (Cash Sale)

Purpose

To process a normal cash sale, update inventory, and generate a receipt.

Algorithm: DailyCashSale

```
BEGIN DailyCashSale
  Initialize SaleSession
  Display Product List

  WHILE Customer has more items DO
    Scan Item
    IF Item exists AND Stock > 0 THEN
      Add Item to SaleSession
      Update Total Amount
    ELSE
      Display "Item not available"
    ENDIF
  ENDWHILE

  Accept Cash Payment
  IF Payment amount >= Total Amount THEN
    Record Sale Transaction
    Update Inventory Stock
    Generate Receipt
    Display "Sale Completed Successfully"
  ELSE
    Display "Insufficient Payment"
  ENDIF
END
```

Figure 21 *DailyCashSale*

Logic Model 2: Credit Sale Processing

Purpose

To process sales made on credit and manage customer credit records.

```
BEGIN CreditSaleProcess
  Input Customer ID or Phone Number
  Search Customer Database

  IF Customer exists THEN
    Retrieve Customer Credit Profile
  ELSE
    Create New Customer Profile
    Assign Unique Customer ID
    Set Initial Credit Limit
  ENDIF

  Calculate Sale Total
  IF Sale Total <= Customer Credit Limit THEN
    Record Credit Sale
    Update Customer Outstanding Balance
    Set Payment Due Date
    Schedule Reminder Notification
    Generate Receipt
  ELSE
    Display "Credit Limit Exceeded"
  ENDIF
END
```

Figure 22 Credit Sale Processing

Logic Model 3: InventoryRestocking

Purpose

To update inventory records when new products are received from suppliers.

Algorithm: InventoryRestocking

```
BEGIN InventoryRestocking
  Receive Products from Supplier
  Input Delivery Details

  FOR each Delivered Item DO
    IF Product exists in system THEN
      Update Stock Quantity
    ELSE
      Create New Product Record
      Set Initial Stock Quantity
    ENDIF
  ENDFOR

  Save Inventory Updates
  Display "Stock Updated Successfully"
END
```

Figure 23 Inventory Restocking

Logic Model 4: Low Stock & Expiry Monitoring

Purpose

To automatically monitor inventory and generate alerts.

Algorithm: StockMonitoring

```
BEGIN StockMonitoring
  FOR each Product in Inventory DO
    IF Stock Quantity <= Minimum Threshold THEN
      Trigger Low Stock Alert
    ENDIF

    IF Expiry Date is Near THEN
      Trigger Expiry Alert
    ENDIF
  ENDFOR
END
```

Figure 24 *StockMonitoring*

Logic Model 5: DemandForecasting

Purpose

To analyze historical sales data and predict future demand.

Algorithm: DemandForecasting

```
BEGIN DemandForecasting
  Collect Historical Sales Data
  Clean and Preprocess Data

  Apply Forecasting Model
  Generate Demand Prediction

  Identify Fast-Moving Products
  Identify Slow-Moving Products

  Store Prediction Results
  Send Insights to Dashboard Module
END
```

Figure 25 *DemandForecasting*

Logic Model 6: End Of Day Closing

Purpose

To finalize daily sales and prepare reports.

Algorithm: End Of Day Closing

```
BEGIN EndOfDayClosing
  Close Sales Register
  Generate Daily Sales Summary
  Count Physical Cash

  IF Physical Cash = System Cash Total THEN
    Mark Day as Balanced
  ELSE
    Record Cash Discrepancy
  ENDIF

  Sync Data with Central System
  Backup Daily Transactions
END
```

Figure 26 *EndOfDayClosing*

2.4 Non-Functional Requirements

Non-functional requirements describe the quality attributes and constraints of the GebayaNet system. These requirements ensure that the system performs efficiently, securely, and reliably in a real business environment.

- The system shall process sales transactions within **2 seconds** under normal operating conditions.
 - The dashboard shall load summarized reports and analytics within **3 seconds**.
 - The system shall support **multiple concurrent users** (manager, cashier, storekeeper) without performance degradation.
 - AI analytics and forecasting processes shall execute in the background without interrupting normal operations.
- Reliability Requirements
- Automatic data backup shall be performed daily to prevent data loss.
 - The system shall recover gracefully from unexpected failures without corrupting stored data.

Security Requirements

- User authentication shall be enforced using **username and password**.
- Role-based access control shall restrict users to authorized functionalities only.
- Sensitive data (user credentials, customer credit information) shall be securely stored.
- Only authorized users (Admin/Manager) shall be allowed to modify system configurations.

Usability Requirements

- The user interface shall be **simple, intuitive, and user-friendly**, suitable for users with limited technical knowledge.
- Common tasks such as sales entry and stock updates shall require **minimal steps**.
- Error messages shall be clear and informative.

Scalability Requirements

- The system shall support future expansion, including additional users, products, and transactions.
- AI models shall be extendable to include more advanced prediction techniques.

2.5 System Requirements

This section specifies the **hardware and software resources** required for deploying and operating the GebayaNet system successfully.

2.5.1 Hardware Requirements

The minimum recommended hardware specifications are as follows:

Server / Hosting Environment

- Processor (CPU): Intel Core i5 (or equivalent) or higher
- RAM: Minimum **8 GB**
- Storage: Minimum **256 GB SSD**
- Internet Connection: Stable broadband connection

Client Devices

- Processor: Dual-core processor or higher
- RAM: Minimum **4 GB**
- Storage: At least **50 GB free disk space**
- Peripherals: o Barcode scanner (optional) o Receipt printer o Keyboard and mouse

2.5.2 Software Requirements

- Server-Side:
 - OS: Windows Server or Linux (Ubuntu 20.04+)
 - Runtime: Node.js v18+ or Python v3.9+
 - Database: MySQL v8.0+ or PostgreSQL
 - Web Server: Apache or Nginx
- Client-Side:
 - OS: Windows 10+, Linux, or macOS
 - Browser: Google Chrome v100+ or Mozilla Firefox v95+
 - Optional Tools: PDF viewer, email API integration software

Conceptual Modeling: Class Diagram

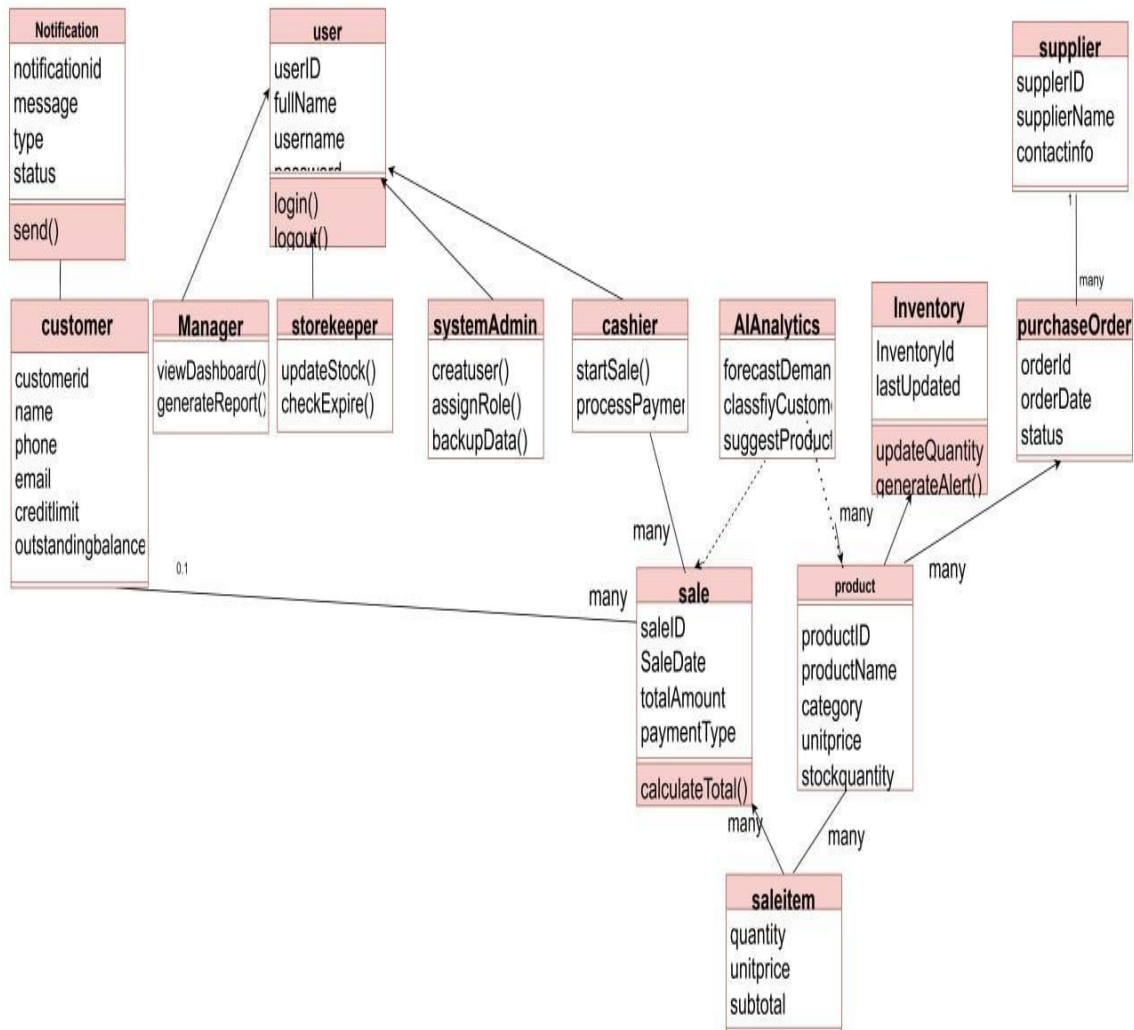


Figure 27 Class Diagram

2.6.1 Identifying Change Cases

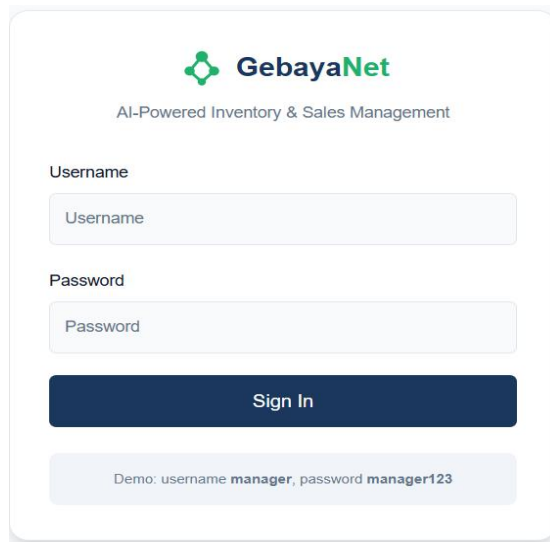
Table 11: Change Cases

Change Case	Description	Likelihood	Impact	Priority
CC-01	Mobile Payment Integration - Add support for local mobile money services (Telebirr, M-Pesa)	High	. Requires payment gateway API integration • Security enhancements needed . Transaction fee management	High
CC-02	Barcode/QR Code Scanning - Implement hardware integration for barcode scanners and QR code readers	High	Hardware interface development • Scanner driver integration . Inventory barcode generation system	Medium
CC-03	Multi-Location Management - Support for multiple store branches with centralized control	Medium	. Database redesign for multi-tenancy . Inter-store transfer tracking . Consolidated reporting	High
CC-04	Advanced AI Analytics Enhanced predictive models with deep learning for better forecasting	Medium	. Requires more historical data . Higher computational resources . Data scientist involvement	Medium
CC-05	Multilingual Support - Add Amharic, Afaan	High	. UI translation . RTL text support for	Medium

	Oromo, and Tigrinya language interfaces		Amharic . Multilingual customer support	
CC-06	Offline Mode - Allow system operation without internet with data sync when connected	Medium	. Local database implementation . Conflict resolution for sync . Data consistency management	Medium
CC-07	Government Tax Integration - Automatic tax calculation and reporting for ERCA compliance	High	. Tax rule configuration • Report generation for authorities . Regular updates for tax law changes	High
CC-08	Supplier Portal - Allow suppliers to view orders and update delivery status	Low	. External user management . Supplier authentication system . Order status tracking portal	Low
CC-09	Customer Mobile App Allow customers to view their purchase history and credit status	Medium	. Separate mobile application . Customer authentication • Privacy and data protection	Medium
CC-10	Inventory Audit Module Automated stock taking with mobile devices	Medium	. Mobile audit app development • Reconciliation algorithms . Variance reporting	Low

2.6.3 User Interface Prototyping

User interface prototyping was used to **visualize system interactions** and validate requirements with users.



The login form for GebayaNet, an AI-Powered Inventory & Sales Management system. It features a green logo with a cross-like symbol. The form includes fields for Username and Password, a dark blue Sign In button, and a demo login link at the bottom: Demo: username **manager**, password **manager123**.

Figure 28 sign

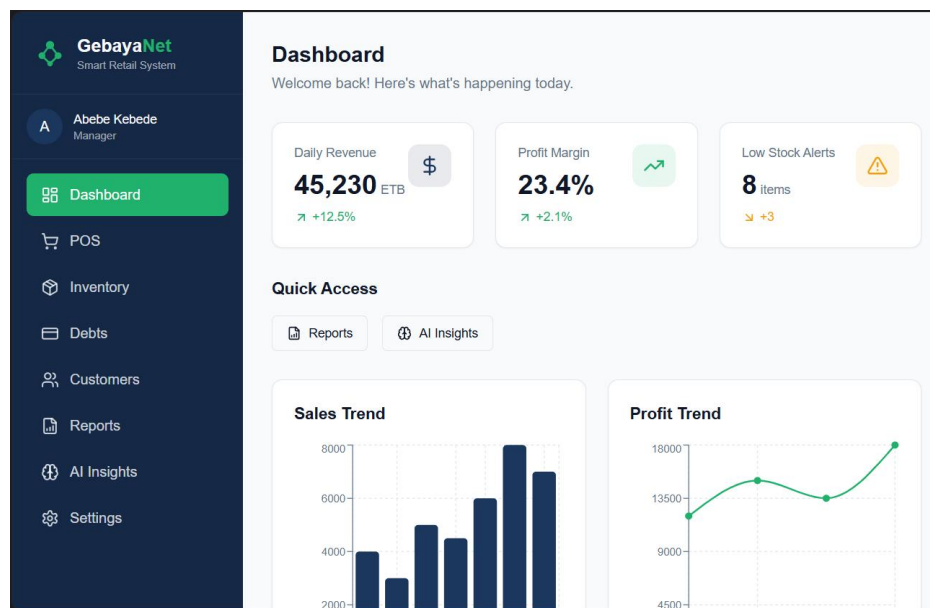


Figure 29 Dashboard UI

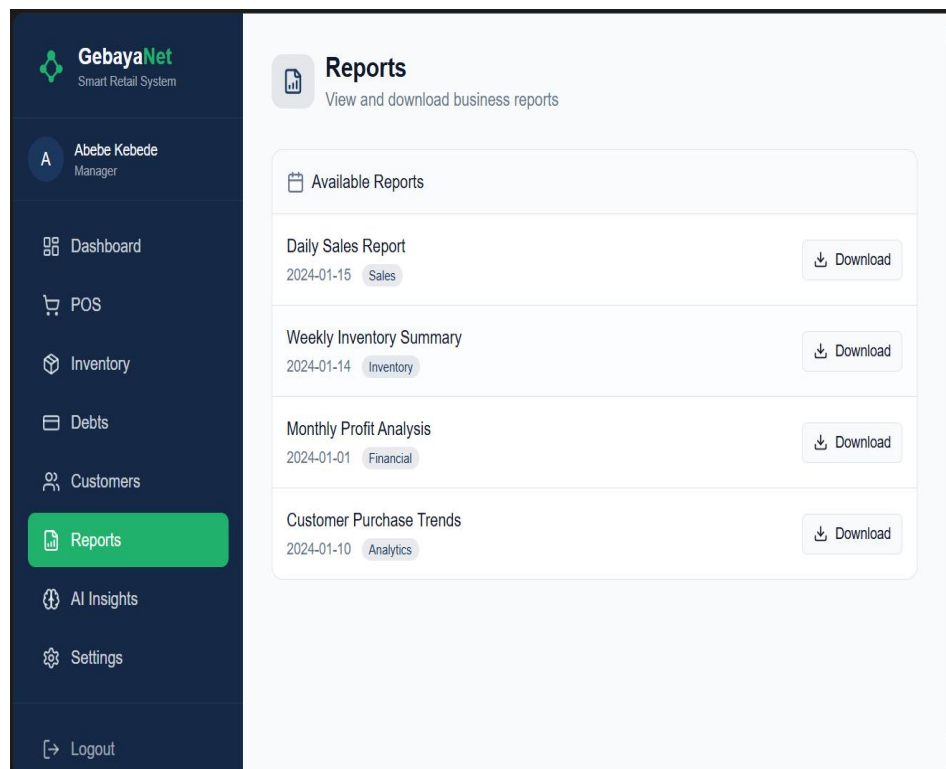


Figure 30 Report UI

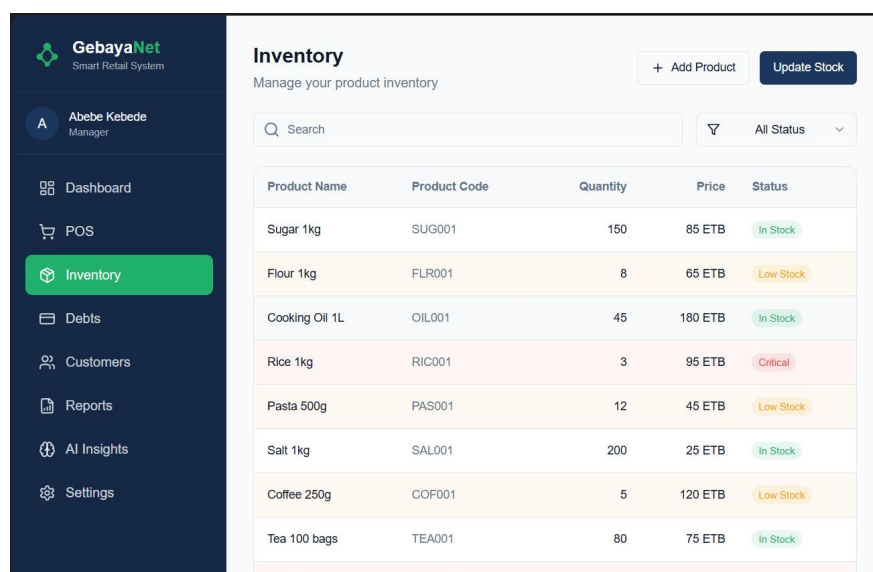


Figure 31 Inventory UI

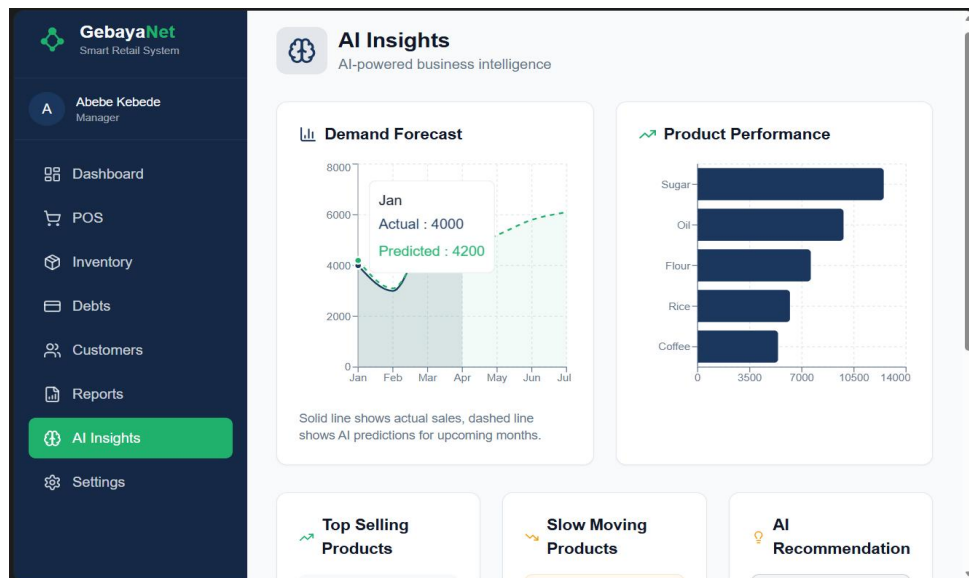


Figure 32 AI Insights UI

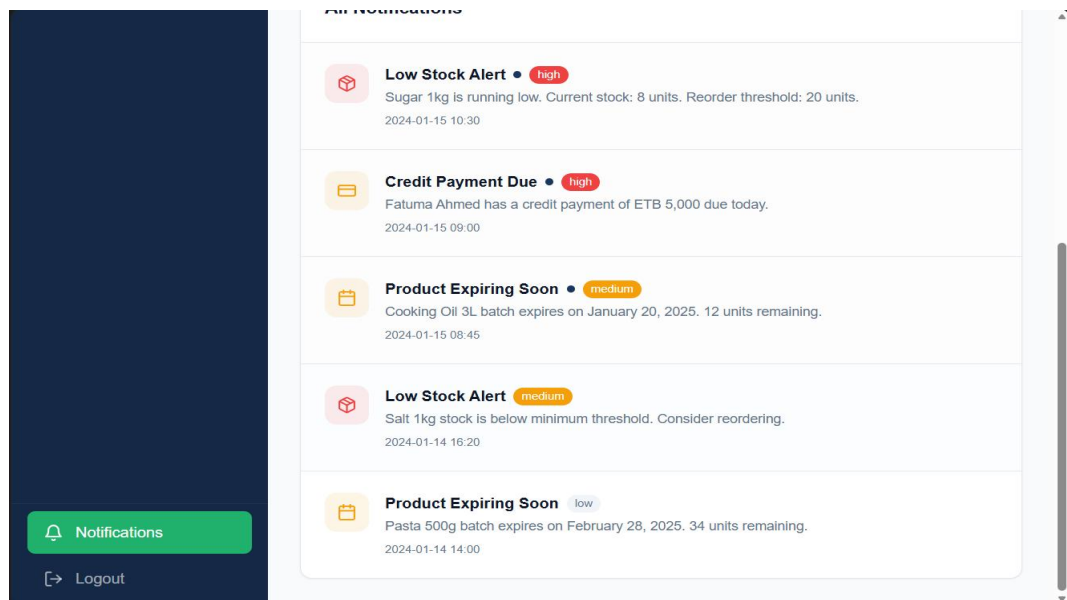


Figure 33 Notification UI

Chapter 3: System Design

This chapter presents the detailed system design for GebayaNet, translating the functional and non-functional requirements established in Chapter 2 into a concrete technical blueprint. The design focuses on achieving key quality attributes: Performance (ensuring responsiveness under typical Ethiopian business loads), Dependability (maintaining high availability and data integrity), Security (protecting business data through role-based access and encryption), Usability (creating intuitive interfaces for users with varying technical skills), and Scalability (allowing for future growth in users, products, and features). The design follows a modular, layered architecture to ensure maintainability and facilitate iterative development.

3.1 Architectural Design

GebayaNet employs a layered microservices architecture that separates concerns and promotes scalability. This design ensures that the system can handle concurrent users typical of a medium-sized Ethiopian retail business while maintaining responsiveness. The architecture is divided into three primary layers: Presentation, Application, and Data



Figure 34 Architectural Design

3.1.1 Component Modeling

The system is decomposed into cohesive, loosely coupled components that interact through well-defined interfaces. This component-based approach enhances maintainability and allows for independent development and scaling of system features.

Frontend Components:

- **Authentication UI:** Handles user login, registration, and password recovery.
- **Manager Dashboard:** Provides real-time analytics, KPIs, and AI insights.
- **POS Interface:** Enables cashiers to process sales quickly and efficiently.
- **Inventory Management:** Allows storekeepers to manage stock levels and view alerts.
- **Admin Panel:** Provides system configuration and user management capabilities.
- **Storekeeper Mobile App:** A React Native companion app for floor operations.

Backend Services (Application Layer):

- **API Gateway:** A single entry point that routes requests to appropriate services, handles load balancing, and enforces security policies.
- **Auth Service:** Manages user authentication, session handling, and token generation (JWT).
- **Sales Service:** Processes all sales transactions (cash/credit), calculates totals, applies taxes, and generates receipts.
- **Inventory Service:** Manages product CRUD operations, tracks stock levels in real-time, and triggers reorder alerts.
- **Reporting Engine:** Aggregates transactional data to generate business intelligence reports and visual dashboards.
- **AI Forecasting Service:** Runs predictive algorithms (exponential smoothing) on historical sales data to generate demand forecasts.

- **Notification Service:** Sends automated alerts for low stock, expiry dates, and overdue credits via in-app notifications.

Data Layer Components:

- **PostgreSQL Database:** Primary relational database for storing structured business data (users, products, sales, inventory).
- **Cloud Storage:** For backup and storing generated reports and system logs.

The components interact through RESTful APIs, ensuring clear separation and allowing for future replacement or enhancement of individual services without affecting the entire system.

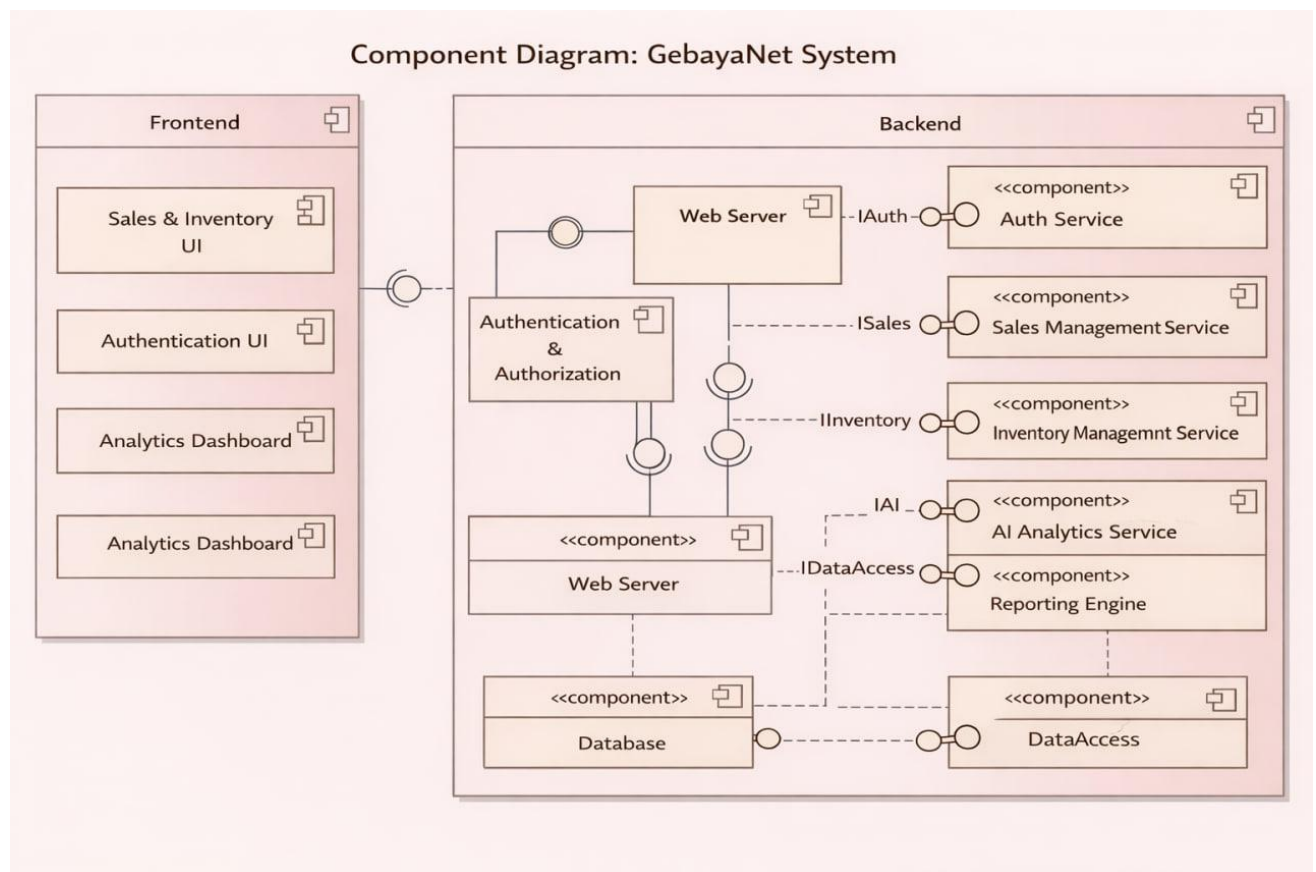


Figure 35 Component Interaction Diagram

3.1.2 Deployment Modeling

GebayaNet will be deployed using a cloud-based infrastructure model to ensure high availability, scalability, and cost-effectiveness, suitable for Ethiopian SME operational environments.

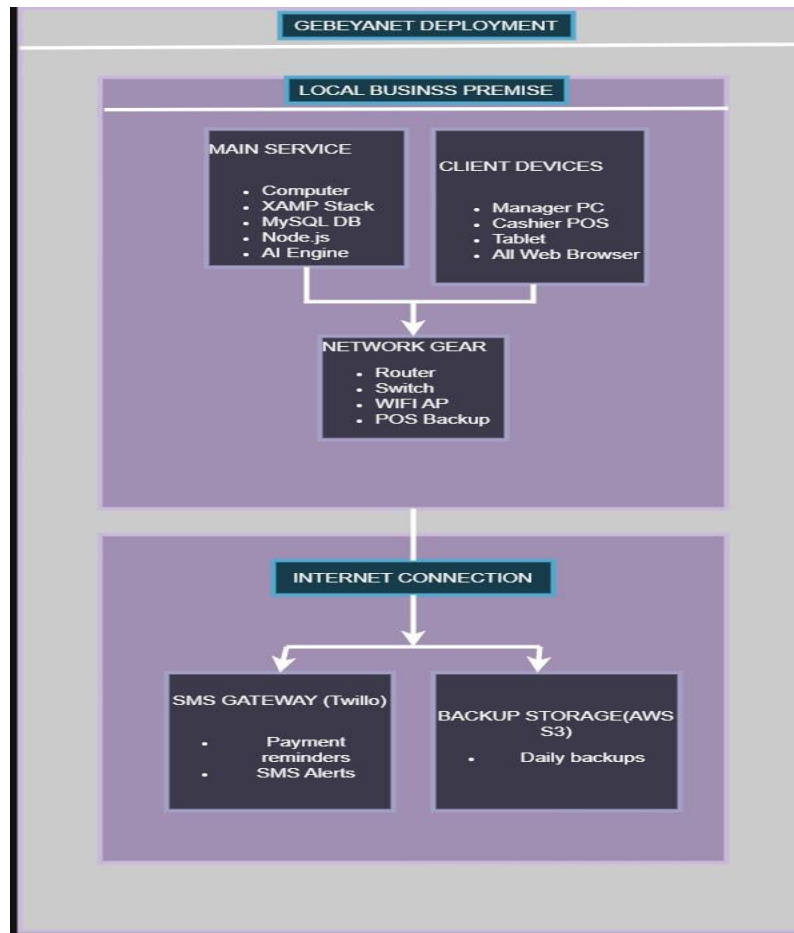


Figure 36 Deployment Diagram

Deployment Nodes and Specifications:

1. Web/Application Server Cluster:

- **Location:** Cloud Virtual Machines (e.g., AWS EC2, DigitalOcean Droplets)
- **Specification:** 2 vCPUs, 8GB RAM, 50GB SSD (per instance)
- **Components Hosted:** Node.js backend services (Auth, Sales, Inventory, Reporting, AI), API Gateway, and Static Frontend files (React build).
- **Load Balancer:** Distributes incoming traffic across multiple server instances for fault tolerance and performance.

2. Database Server:

- **Location:** Managed Database Service (e.g., AWS RDS for PostgreSQL, DigitalOcean Managed Databases)
- **Specification:** 2 vCPUs, 8GB RAM, 200GB SSD Storage with automated backups.
- **Components Hosted:** Primary PostgreSQL database containing all business data.

3. Caching Server:

- **Location:** Managed In-Memory Cache (e.g., AWS ElastiCache, Redis Cloud)
- **Specification:** 1 vCPU, 2GB RAM.
- **Components Hosted:** Redis instance for session storage and data caching.

4. Client Devices:

- **Location:** On-premise at the business location (shop, warehouse).
- **Hardware:** Desktop PCs (for manager/admin), Point-of-Sale terminals or tablets (for cashiers), Android tablets/smartphones (for storekeeper mobile app).

- **Connectivity:** Requires stable internet connection (≥ 5 Mbps) for real-time synchronization. Basic offline transaction recording is supported on mobile devices with sync upon reconnection.

Communication Protocols:

- **Frontend-Backend:** HTTPS (TLS 1.3) for secure API communication.
- **Backend-Database:** Secure connection over TLS using client-certificate authentication.
- **Server-Server:** Internal REST API calls within a private cloud network.

3.2 Detail Design

3.2.1 Design Class Model

The Design Class Model refines the analysis classes by adding implementation-specific details such as method signatures, return types, and visibility modifiers. It defines the core business logic structure of the GebayaNet system.

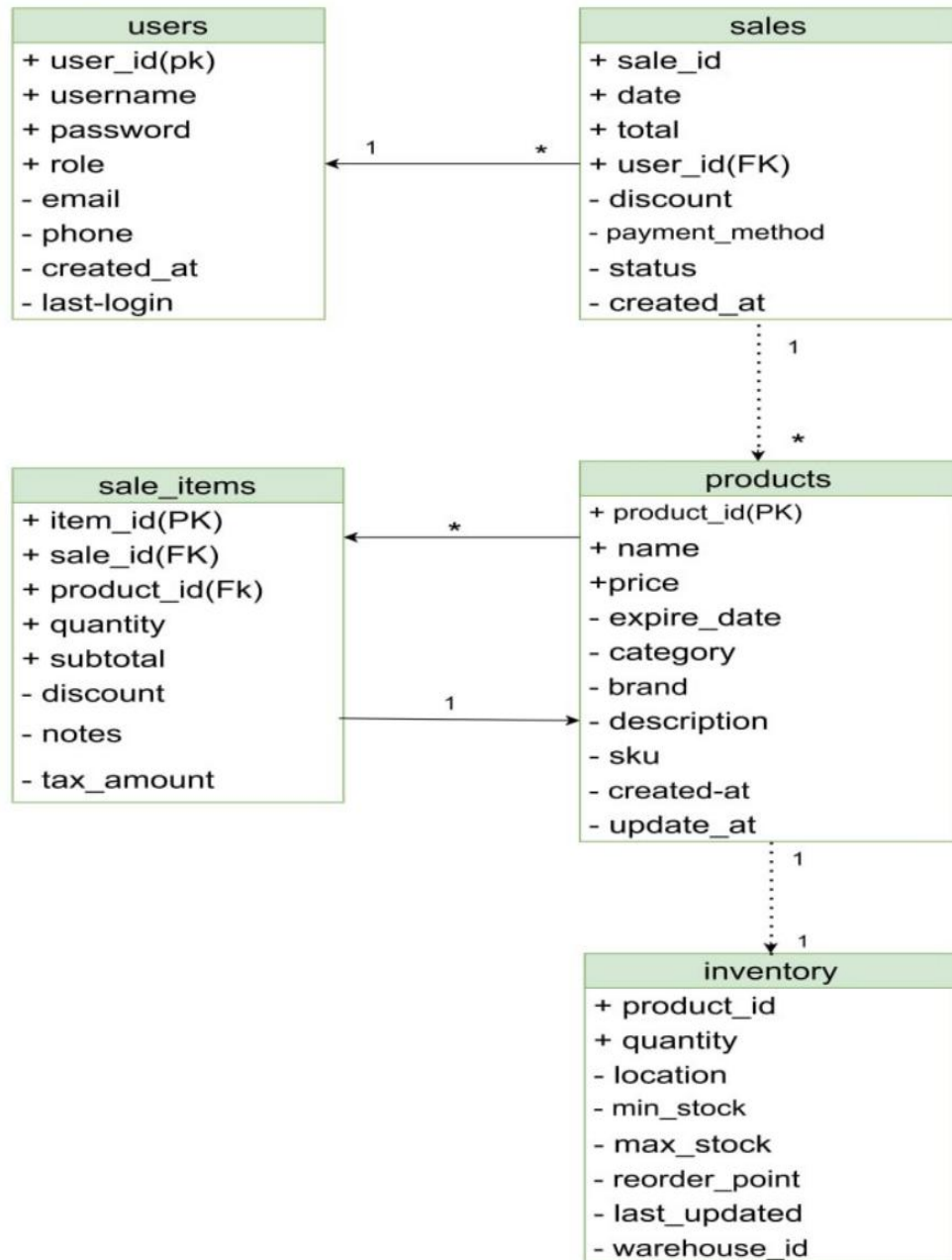


Figure 37 Design Class Model

3.2.2 Persistent Model

The persistent model defines the relational database schema that will store all system data. It is designed for data integrity, efficient querying, and scalability.

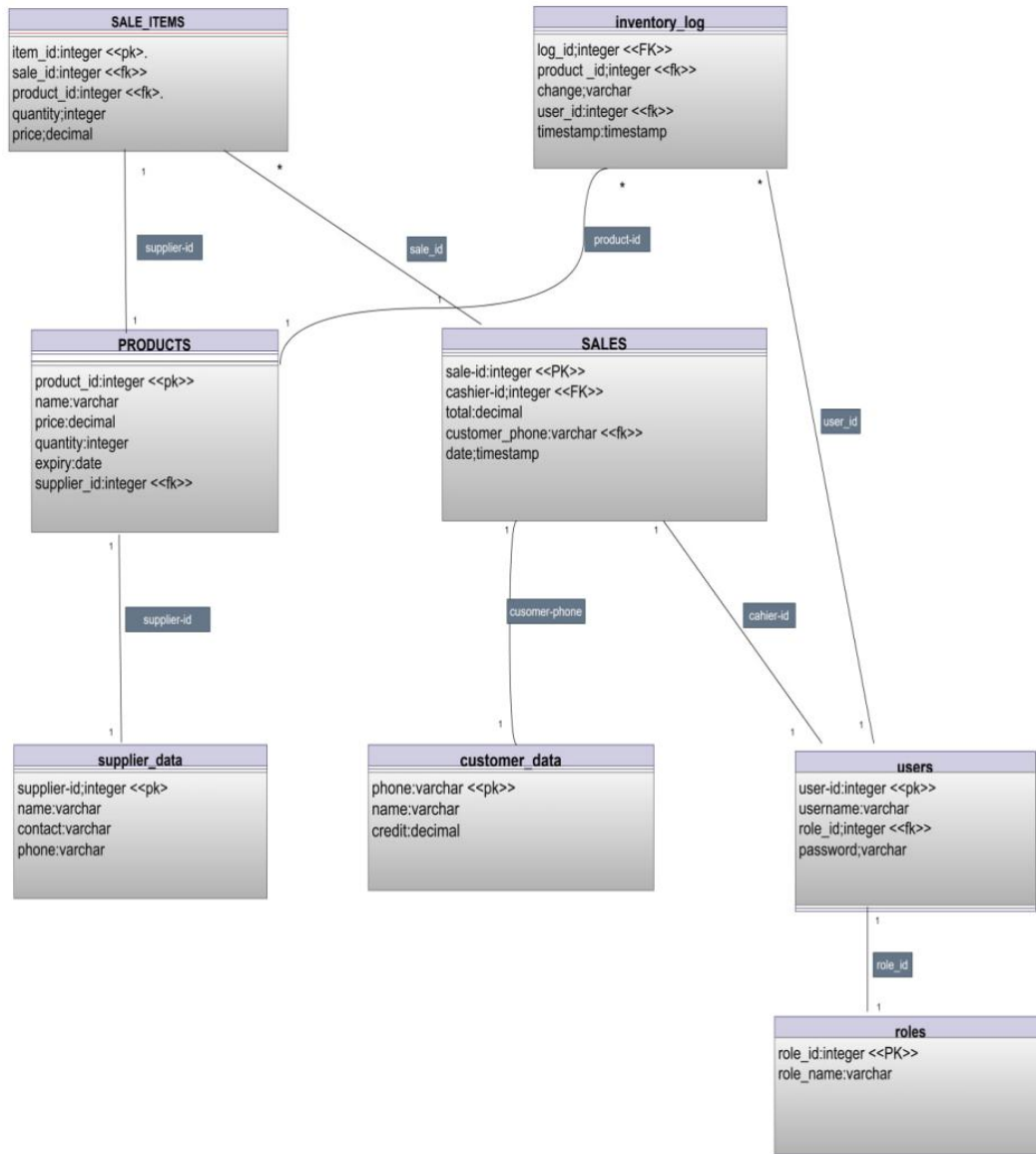


Figure 38 Persistent Model Diagram

Here are sample JSON structures for the key persistent model entities in GebeyaNet:

a. User

```
{
  "user_id": 1001,
  "username": "manager_kebede",
  "password": "$2b$10$hashedpasswordexample",
  "role": "manager",
  "email": "kebede@shop.com",
  "phone": "+251911234567",
  "created_at": "2023-01-15T09:30:00Z",
  "last_login": "2024-06-10T14:25:00Z"
}
```

Figure 39 User

b. Product

```
{
  "product_id": 5001,
  "name": "Coca-Cola 500ml",
  "sku": "BEV-CC-500",
  "category": "Beverages",
  "brand": "Coca-Cola",
  "unit_price": 25.50,
  "expiry_date": "2024-12-01",
  "description": "Carbonated soft drink",
  "created_at": "2024-01-10T08:00:00Z",
  "updated_at": "2024-06-01T10:15:00Z"
}
```

Figure 40 Product

c. Inventory

```
{
  "inventory_id": 7001,
  "product_id": 5001,
  "current_quantity": 120,
  "min_stock": 50,
  "max_stock": 500,
  "reorder_point": 100,
  "location": "Shelf A-12",
  "last_updated": "2024-06-10T16:45:00Z",
  "warehouse_id": "WH-01"
}
```

Figure 41 Inventory

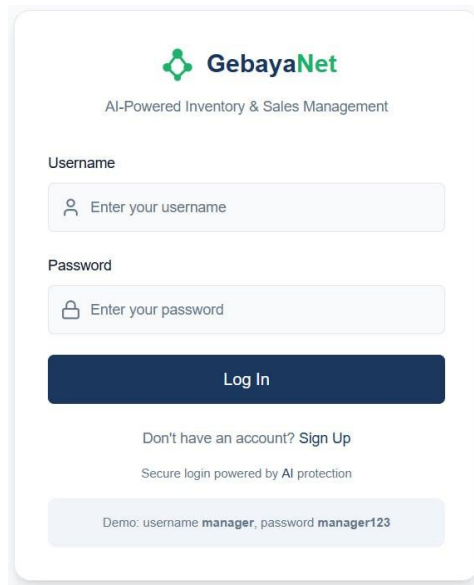
e. Forecast Result

```
{
  "forecast_id": "FC-202406-001",
  "product_id": 5001,
  "forecast_date": "2024-06-01",
  "period": "weekly",
  "predicted_quantity": 145,
  "confidence_interval": 0.85,
  "algorithm_used": "exponential_smoothing",
  "generated_at": "2024-06-01T23:59:00Z"
}
```

Figure 42 Forecast Result

3.3 User Interface Design

The UI design prioritizes simplicity, clarity, and efficiency to accommodate users with varying levels of technical expertise. It follows a consistent design system with a neutral color scheme, clear typography, and intuitive iconography.



The login form for GebayaNet features a clean, modern design. At the top, the GebayaNet logo is displayed alongside the tagline 'AI-Powered Inventory & Sales Management'. Below this, there are two input fields: 'Username' with a placeholder 'Enter your username' and 'Password' with a placeholder 'Enter your password'. A dark blue 'Log In' button is positioned below the password field. Underneath the button, there is a link for users who don't have an account to 'Sign Up'. A security note states 'Secure login powered by AI protection'. At the bottom, a demo login credential is provided: 'Demo: username manager, password manager123'.

Figure 43 **Login**

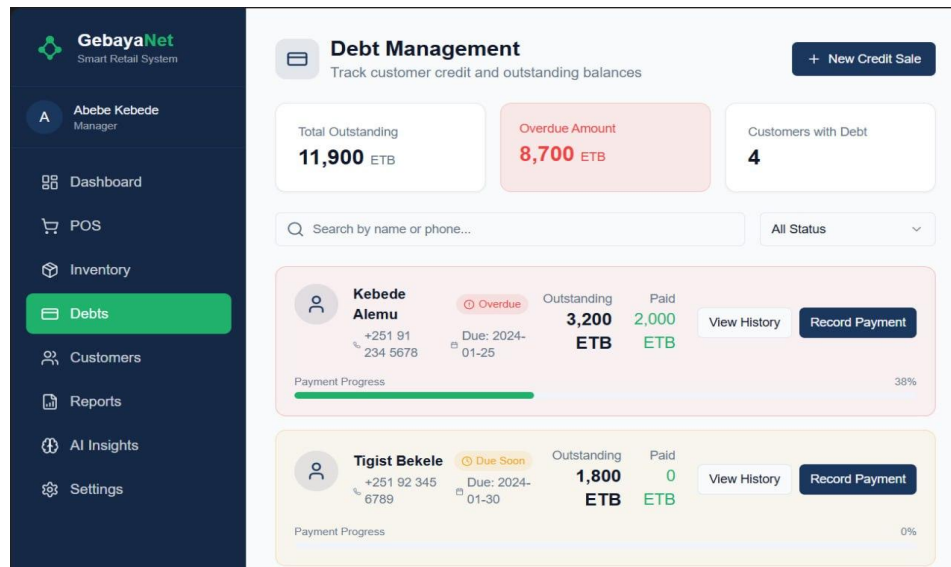


Figure 44 **debts**

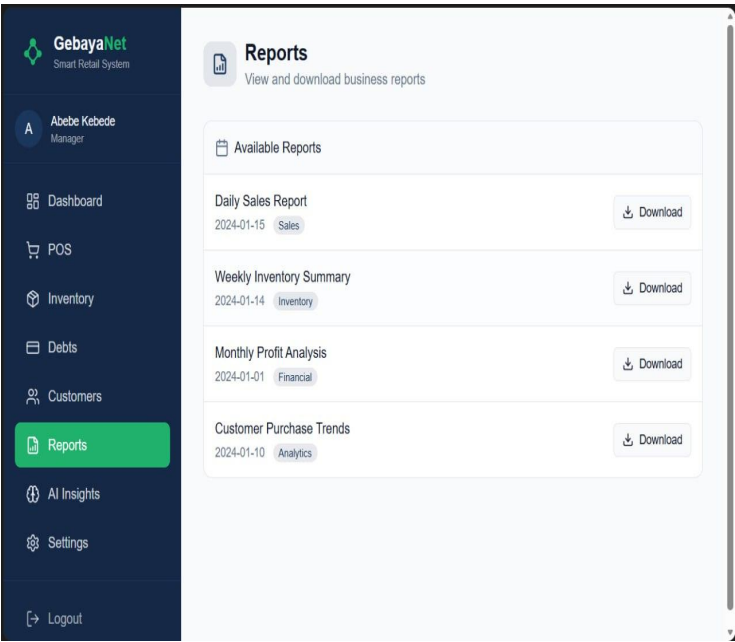


Figure 45 Reports

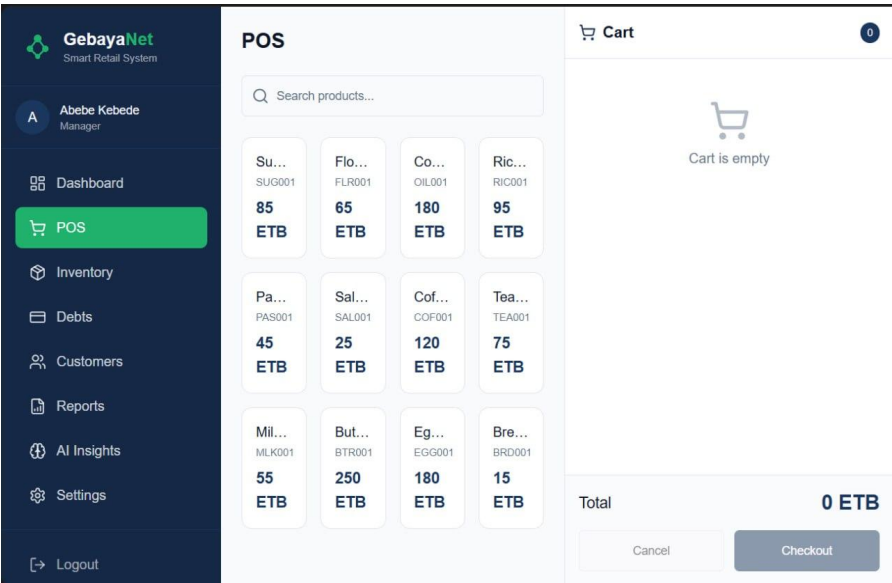


Figure 46 POS

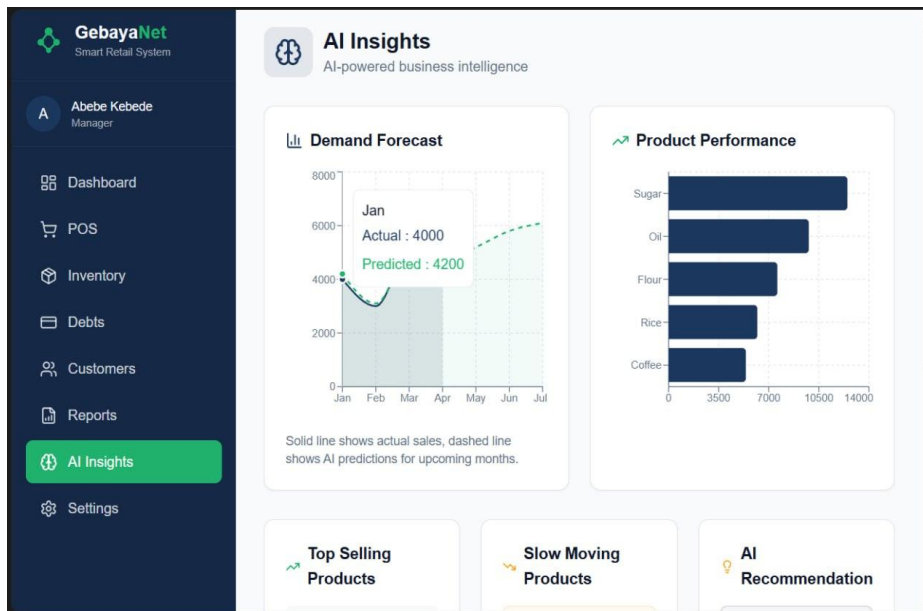


Figure 47 **AI Insights**

GebayaNet
Smart Retail System

Abebe Kebede
Manager

- Dashboard
- POS
- Inventory
- Debts
- Customers
- Reports
- AI Insights
- Settings**

Settings

Manage your account and preferences

Profile Settings

First Name: Abebe

Last Name: Kebede

Email: abebe@example.com

Store Settings

Store Name: Kebede General Store

Language: English

Figure 48 **Settings**

Design Principles:

1. **Role-Based Layout:** Each user role (Manager, Cashier, Storekeeper) has a customized dashboard highlighting relevant tasks and information.
2. **Progressive Disclosure:** Complex features (like AI forecasting parameters) are hidden behind simple controls, revealing advanced options only when needed.
3. **Action-Oriented Navigation:** Primary actions (e.g., "New Sale", "Add Product") are prominently placed and accessible within one click.
4. **Responsive Design:** The web interface adapts to different screen sizes, from desktop monitors to tablets used at the POS.

3.4 Access Control and Security

Security is paramount for a system handling sensitive business and financial data. GebayaNet implements a multi-layered security strategy.

1. Authentication & Authorization:

- **Authentication:** Users must log in with a unique username and a strong, hashed password (using bcrypt).
- **Authorization:** A **Role-Based Access Control (RBAC)** system is strictly enforced, as defined in Business Rules BR-01 and BR-02. Permissions are granular and defined per role:
 - **Manager:** Read/write access to all modules except user/role administration. Can view all reports and AI insights.
 - **Cashier:** Can process sales (cash/credit) and view product catalog. *Cannot* modify product prices, manage users, or view financial reports.
 - **Storekeeper:** Can update inventory levels, add/edit product information, and view stock alerts. *Cannot* process sales or view sales reports.

- **System Administrator:** Full system access, including user management, role assignment, and system configuration. Does not typically engage in daily business operations.

2. Data Security:

- **Encryption:** Sensitive data at rest (customer credit info, user passwords) is encrypted using AES-256. All data in transit is protected via TLS/SSL.
- **Input Validation & Sanitization:** All user inputs are rigorously validated on both client and server sides to prevent SQL injection and cross-site scripting (XSS) attacks.
- **Audit Logging:** All critical actions (user login, sale deletion, inventory override) are logged with timestamp and user ID for non-repudiation and security auditing.

3. Session Management:

- Sessions are managed using secure, HTTP-only cookies with JWT tokens.
- Automatic session timeout is enforced after 30 minutes of inactivity.
- Concurrent login from multiple devices for the same user is configurable by the administrator.

4. Data Backup and Recovery:

- As per BR-12, automated daily backups of the database are performed to a secure, geographically separate cloud storage location.
- Backup integrity is verified regularly, and a documented disaster recovery procedure ensures business continuity in case of system failure.

Conclusion

This project focused on the analysis and design of GebayaNet, an inventory and sales management system intended to support the daily operations of small and medium retail businesses. The system was designed to address common challenges such as manual stock tracking, inefficient sales and credit recording, and limited access to meaningful business information. By combining inventory control, sales processing, reporting, and basic AI-based demand forecasting, GebayaNet offers a practical and organized approach to managing business activities.

Throughout the project, functional and non-functional requirements were carefully identified and modeled using object-oriented analysis and UML diagrams. The proposed system architecture emphasizes simplicity, scalability, and security while remaining feasible within the technical and financial constraints of local businesses. The use of accessible technologies and free or low-cost deployment options makes the system realistic for real-world implementation.

In conclusion, this project provides a clear and structured foundation for developing a functional inventory and sales management system. While further development, testing, and real-world usage are required to fully evaluate its effectiveness, the proposed design demonstrates how digital tools can gradually improve record-keeping, decision-making, and overall business management for Ethiopian retailers.

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